

Cybersecurity & Cyber-Resilience Strategy for the Sovereign Cloud–Edge–IoT Continuum

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Abstract

The rapid decentralization of computing infrastructures toward the Cloud–Edge–IoT continuum introduces unprecedented cybersecurity challenges, including expanded attack surfaces, heterogeneous environments, and increased dependency on non-sovereign technologies. This paper proposes a structured, multi-stage methodology for defining and prioritizing cybersecurity capabilities required to achieve a secure and sovereign European computing continuum. The approach combines a pipeline-oriented transformation model with a seven-dimensional analytical framework, enabling traceable transitions from architectural capabilities to strategic priorities and policy-aligned recommendations. A four-filter prioritization logic identifies critical capabilities based on regulatory urgency, architectural dependencies, sovereignty risks, and market feasibility. These priorities are further analyzed through a progression matrix capturing operational deployment and strategic sovereignty evolution across time phases. The methodology provides a reproducible and evidence-based framework for guiding research, innovation investments, and policy decisions in the context of the European digital sovereignty agenda.

Keywords: *Cybersecurity, Cloud-Edge-IoT, Digital Sovereignty, Zero Trust, Post-Quantum Cryptography, EU Policy, Cyber-Resilience*

1 Introduction

1.1 Contributions

This paper makes the following contributions:

- A structured pipeline methodology for cybersecurity capability prioritization in the Cloud–Edge–IoT continuum.
- A 4-dimensional filtering logic linking regulatory, architectural, sovereignty, and market-adoption considerations.
- A hybrid 7-dimensional profiling framework for actionable R&I prioritization.
- A progression matrix model capturing operational deployment and strategic sovereignty evolution across time phases.
- A traceable transformation from technical capabilities to policy-aligned strategic recommendations.

This manuscript and roadmap defines the architectural guardrails required to secure the Cloud–Edge–IoT Continuum. It moves beyond traditional “perimeter security” toward a model of cyber-resilience—where systems are engineered to proactively anticipate, withstand, recover from, and adapt to attacks while maintaining strict digital sovereignty. **Approach:** To provide actionable, risk-aware Research & Innovation (R&I) priorities, the strategy is organized into a two-part framework:

Purpose of This Document

This roadmap provides a structured methodology for European policymakers, research organizations, and industry stakeholders to:

- Identify and prioritize cybersecurity R&I investments for the Cloud-Edge-IoT continuum
- Track capability maturity using objective Operational Deployment Phase (ODP) and Strategic Sovereignty Level (SSL) metrics.
- Align Horizon Europe funding with strategic autonomy goals and regulatory compliance requirements
- Ensure coordinated implementation across Member States and European initiatives (ECCC, Digital Europe Programme, EU Chips Act)

Target Audiences

European Commission & Funding Bodies: Use the 4-Filter Prioritization Framework to allocate Horizon Europe, Digital Europe Programme, and ECCC funding to capabilities with the highest strategic impact.

National Cybersecurity Agencies: Leverage the progression matrices to coordinate national R&I investments with EU-wide timelines and ensure interoperability across Member States.

Research Consortia & Industry: Utilize the 7-Dimensional Analysis template to structure project proposals, clearly articulating baseline, target, drivers, risks, and alignment with European strategic goals.

Standardization Bodies (CEN/CENELEC, ETSI): Reference the architectural framework and MIMs to develop harmonized standards for secure, interoperable computing continuum infrastructure.

How to Navigate This Document

1. **Start with Section 1-3:** Understand the threat landscape, regulatory drivers, and strategic context for securing the European computing continuum.
2. **Review Section 4:** Familiarize yourself with the 6 Security Layers and 4 Cross-Layers comprising the architectural framework.
3. **Apply Section 5:** Use the 4-Filter Prioritization Framework to assess which capabilities are critical for your specific context (national strategy, industrial sector, research focus).

4. **Study Section 6-7:** Understand the 7-Dimensional Analysis methodology and ODP/SSL progression metrics before designing projects or policies.
5. **Reference Section 9 (Priorities 1-3):** Use the detailed capability profiles as templates for structuring additional priorities relevant to your domain.
6. **Implement Section 10:** Actionable strategic recommendations for Phase 1 (2025-2030) provide concrete next steps.

Key Principles

- **Evidence-Based Prioritization:** All priorities must pass the 4-Filter Framework (Regulatory Urgency, Foundational Dependency, Sovereignty Cliff, Market Feasibility).
- **Sovereignty-First Design:** Investments should systematically reduce foreign dependencies (measured via SSL metrics) while building European industrial capacity.
- **Compliance-by-Design:** All R&I outputs must align with or anticipate NIS2, CRA, AI Act, and Data Act requirements to ensure market relevance.
- **Federated Interoperability:** Solutions must prioritize standardization and open architectures (e.g., Simpl middleware, open-source RISC-V) over proprietary, siloed approaches.
- **Measurable Progression:** Avoid subjective TRL assessments; instead, use objective ODP (market integration) and SSL (sovereignty) metrics with clear phase transition criteria.

2 Background & Driving Factors

Destinations:

- **Primary:** 4. A secure, sovereign European computing continuum infrastructure.
- **Secondary:** 6. European leadership in emerging disruptive computing paradigms.
- **Decentralization Risk:** The paradigm shift from the centralized Cloud to a distributed Cloud-Edge-IoT continuum radically expands the attack surface. This creates environments that are often “too many, too diverse, and too dispersed” to secure with traditional perimeter defenses.
- **Physical Exposure:** Edge nodes and far-edge IoT devices frequently operate in the field. Unlike hyperscaler data centers, they lack strict physical and environmental security controls, making them highly vulnerable to tampering.
- **Regulatory Pressure:** There is an immediate, complex need to operationalize the NIS2 Directive [1], the Cyber Resilience Act (CRA) [2], and the AI Act [3] across a highly fragmented technological ecosystem to ensure compliance and market access.
- **Digital Sovereignty:** Europe must urgently reduce its systemic reliance on non-EU technology providers. Doing so will mitigate extra-territorial data access risks and avoid geopolitical vendor lock-in, ensuring independent control over critical infrastructure.

2.1 Europe in Relation to the State of the Art

Comparison vs. Hyperscalers:

- **EU Strengths:** Europe leads globally in Trust, Privacy-by-Design, proactive regulation (GDPR, Data Act, AI Act), and rigorous, standardized certification frameworks (e.g., EUCS [4]).
- **EU Gaps:** The European market suffers from a lack of unified, hyper-scale security orchestration. Furthermore, there is a critical dependency on non-EU hardware (secure silicon) and highly fragmented threat intelligence sharing among member states.

Strategic Goal: To definitively transition the European computing continuum from a state of “reactive, siloed security” to an advanced architecture defined by “autonomous, federated, and certified Security-by-Design.”

3 Related Work

Existing research on cybersecurity for distributed computing infrastructures has primarily focused on isolated domains such as cloud security, edge computing protection, and IoT security frameworks. However, limited work addresses the holistic security challenges of the integrated Cloud–Edge–IoT continuum.

Recent initiatives, including the European Union Cybersecurity Strategy, ENISA reports, and Horizon Europe projects, emphasize the need for sovereignty-driven security architectures. Prior studies on Zero Trust Architectures (ZTA), Post-Quantum Cryptography (PQC), and confidential computing provide important building blocks but lack a unified prioritization and transition methodology.

This work differentiates itself by proposing a structured, end-to-end methodological pipeline that connects architectural capabilities with policy-driven prioritization and phased implementation strategies.

4 Methodology Overview

4.1 Master Capability Transition Framework - Transition Matrix

The ten executive priorities identified through the 4-Filter Prioritization Framework in Section 7 do not mature independently or at uniform rates. Different priorities enter Phase 1 at distinct Operational Deployment Phase (ODP) levels: Priorities 3, 6, 8, and 9 commence at the Applied R&D stage because their enabling technologies are not deployable as of 2025–2026, whereas Priorities 1, 2, 4, 5, 7, and 10 commence at Fragmented Piloting under regulatory activation. This heterogeneity is a structural feature of the roadmap and must be preserved across all per-priority analyses; collapsing it to a single Phase 1 characterisation would constitute an overclaim of maturity for the less advanced priorities.

To provide a coherent methodology for tracking capability progression across all ten priorities simultaneously, this section introduces a **Master Capability Transition Framework** structured as a five-row, four-column matrix. This framework is the primary analytical instrument of the roadmap and constitutes the reference architecture against which all per-priority progression matrices are instantiated. The matrix is not a summary of those per-priority matrices: it defines the *transition logic* governing progression through the roadmap, including the explicit gate conditions that must be satisfied before any priority can advance from one phase to the next.

Y-Axis: Phase States and Gate Conditions

The Y-axis defines five states of capability maturity. The first state (Current Status) establishes the 2025 baseline across all analytical dimensions. Three sequential phase states follow, corresponding to the roadmap’s temporal structure: Phase 1 (2025–2030, Foundation and Compliance), Phase 2 (2030–2035, Industrial Scaling, Standardisation and Cross-Sector Deployment), and Phase 3 (2035–2040, Autonomous Leadership and Frontier Innovation). The fifth state (Strategic End-State Objective, 2040 Vision) describes the target condition that the preceding three phases collectively work to achieve.

Between each consecutive row pair, an explicit **gate condition** is defined. A gate condition is the named, verifiable event — a specific regulatory enforcement date, a named standard finalization, or a measured technical feasibility threshold — whose activation is the necessary condition for transition to the subsequent phase. Gate conditions are derived exclusively from the Critical Enabler cells of the ten per-priority progression matrices and are not aspirational descriptions. If a gate condition has not been activated, the corresponding priority has not transitioned, regardless of elapsed calendar time.

Each phase row is annotated with a **Strategic Sovereignty Level (SSL)** band indicating the expected foreign-dependency trajectory during that phase. The SSL band is a row-level property rather than a full column because sovereignty evolves at a rate that is largely co-determined by the four analytical dimensions simultaneously; representing it as a separate column would create structural redundancy with the per-priority ODP/SSL matrices. The SSL progression across the five rows is: High Foreign Dependency (Current Status and early Phase 1) → Emerging EU Ecosystem (late Phase 1 into Phase 2) → Full Sovereign Autonomy (Phase 3) → Full Sovereign Autonomy with Export Leadership (Strategic End-State).

X-Axis: Four Analytical Dimensions

The X-axis defines four analytical dimensions capturing distinct transition logics that operate simultaneously across each phase. **Research and Innovation Maturity** tracks the scientific and technological advancement trajectory, from academic fragmentation through applied and industrial R&D to frontier research, using TRL stage as a calibration anchor. **Policy, Regulatory and Standards Enablement** tracks the activation of the EU regulatory stack — CRA, NIS2, AI Act, DORA — and the maturation of standardisation outputs from ETSI, CEN/CENELEC, 3GPP, and NIST, which function as gate conditions for deployment transitions. **Operational Deployment and Adoption** tracks observable deployment reality: what is actually deployed, by which actor categories (national agencies, critical infrastructure operators, research consortia, SMEs), and at what geographic and sectoral scale. **Market Uptake and Strategic Impact** tracks EU market share in critical security components, SME participation economics, competitive positioning against non-EU hyperscalers, and the strategic autonomy consequences of those market dynamics. This column is explicitly not a duplicate of the Operational dimension: a capability can be operationally deployed while remaining commercially inaccessible to SMEs or strategically dependent on non-EU supply chains, as the case of Intel-TDX-based confidential computing in Phase 1 illustrates.

Row V: Triple Accountability Framework

The Strategic End-State Objective row is accountable to three distinct frameworks simultaneously and must not be reduced to any single one of them. First, **SNS JU SRIA alignment**: Priorities 6, 7, 9, and 10 must satisfy the SNS JU SRIA Phase 3 security targets for 6G and the computing continuum [5]. Second, **ECCC Strategic Agenda alignment**: all ten priorities must contribute to the ECCC’s mandate for European industrial cybersecurity competitiveness and reduced foreign dependency, measured through SSL metrics [6]. Third, **EU regulatory stack coherence**: the full suite of NIS2, CRA, AI Act, DORA, Data Act, EU Cloud Rulebook, and the envisioned “Schengen for Cloud” framework must operate as a coherent, machine-enforceable system by 2040, with no compliance gap between deployed sovereign capability and legal obligation [1–3].

Table 1 presents the full Master Capability Transition Framework. Key performance indicators cited in individual cells are derived directly from the per-priority progression matrices in Section 7 and are not independently generated; they serve as cross-reference anchors enabling consistency verification between this master framework and the detailed priority profiles.

Table 1: Master Capability Transition Framework: Phase States, Gate Conditions, and SSL Bands across Four Analytical Dimensions

Phase / SSL Band	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Impact	
I. Current Status <i>2025 baseline</i> SSL: High F	Fragmented, academically dominated. EU R&I outputs concentrated in Horizon 2020 consortia; sovereign alternatives at TRL 3–5. Heavy dependency on US-led outputs: NIST PQC process, Intel TDX/AMD SEV-SNP TEE architectures, US/Chinese foundation models for SecOps. No coordinated EU-wide applied R&I programme for sovereign secure silicon or edge-native confidential computing.	Regulatory stack adopted but not enforced. NIS2 enforcement initiated (Oct. 2024); CRA implementing technical acts pending; AI Act adopted with high-risk provisions not yet operational; EUCS candidate scheme non-binding; EUCC operational but limited uptake; ENISA PQC migration guidance absent; NIS2 transposition fragmented across Member States.	No continental sovereign security infrastructure. TEE deployment limited to foreign hyperscaler implementations; PQC confined to experimental cloud pilots; Simpl pre-launch with no operational security MIMs; federated SOC architecture at bilateral proof-of-concept stage only; Island Mode not standardised; supply chain attestation manual and opaque in Open RAN ecosystems.	Near-zero EU market share in critical security components. EU domestic production of critical secure silicon near 0%; US/Asian hyperscalers dominate cloud security tooling and foundation models; no EU-sovereign LLM for SecOps; SME compliance costs for CRA/NIS2 unaddressed by support schemes; RISC-V commercial security products absent from public procurement frameworks.	
⇒ Gate condition I→II: R&I: Horizon Europe calls for sovereign secure silicon and lightweight PQC issued; ECCC work programmes activated. Policy: CRA implementing technical acts in force; NIS2 Art. 21 national implementing guidance published by competent authorities. Operational: At least one national pilot per priority domain initiated under regulatory pull. Market: ECCC-co-funded SME compliance support scheme operational; IPCEI-CIS initial deployments contracted.					

Phase / SSL Band	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Impact
II. Phase 1 <i>2025–2030</i> <i>Pilot Validation & Applied Maturity</i> SSL: High	Applied R&D (P3,P6,P8,P9); Fragmented Piloting (P1,P2,P4,P5,P6) Horizon Europe calls targeting sovereign secure silicon, lightweight PQC for 8/16-bit MCUs, Simpl security MIMs, and LLMSecOps. RISC-V secure element prototypes targeting TRL 6–7 under EU Chips Act co-funding. RF fingerprinting and CSI-based authentication validated on EU-manufactured constrained hardware [7]. ECCC-funded crypto-agility middleware research initiated per FIPS 203/204/205 adoption [6]. <i>KPI: RISC-V secure element prototypes at TRL 6–7; lightweight PQC benchmarked on ARM Cortex-M4 class hardware.</i>	Regulatory activation creates compliance market pull. CRA hardware security implementing rules published; NIS2 Art. 21 implementing guidance issued; ENISA PQC sector-specific migration guidelines published by 2026, mapping FIPS 203/204/205 to NIS2 obligations [1]; EU AI Act Art. 9 compliance frameworks operational for high-risk security AI [3]; EUCS certification rollout mandating TEE workload protection [4]; Simpl-Open official launch with initial security MIM definitions; SBOM templates standardised and mandated in EU-funded deployments from 2026 [2]. <i>KPI: ENISA PQC guidelines published and adopted by 5+ Member States by 2026.</i>	Pilot deployments across priority domains under regulatory pull. 3–5 RISC-V secure element pilots under Horizon Europe targeting EUCC ‘High’; LLM-assisted SOC pilots in 5+ national cybersecurity centres under AI Act Art. 9; CAMARA API profiles deployed in 3+ cross-border 5G corridor pilots by 2027; federated SIEMs with MITRE ATT&CK integration operational; Simpl security pilots in 2+ Common Data Spaces (Manufacturing, Health); Island Mode fallback tested in NIS2/DORA-compliant edge architectures [8]. <i>KPI: 2+ Common Data Spaces piloting Simpl security MIMs; federated SIEMs active in 3+ Member States.</i>	Sovereignty trajectory established; SME access initiated. SME compliance support scheme co-funded by ECCC and national agencies, providing subsidised tooling and certification assistance; IPCEI-CIS initial deployments activating bilateral operator interconnection; EU secure silicon market share below 10% but industrial investment trajectory established under CRA pull; EUCC ‘High’ referenced in public procurement frameworks for critical infrastructure ICT. <i>KPI: EU secure silicon market share <10% but investment trajectory established; SME support scheme operational in 15+ Member States.</i>
⇒ Gate condition II → III: R&I: RISC-V secure element prototypes validated at TRL 6–7; lightweight PQC performance benchmarked on EU-manufactured constrained hardware. Policy: ENISA PQC guidelines published; EUCS certification rollout active; Simpl-Open launched with security MIM definitions. Operational: Federated SIEM pilots with standardised CTI sharing operational in 3+ Member States; 2+ Common Data Space Simpl pilots validated. Market: CRA enforcement generating measurable industrial investment in sovereign hardware alternatives.				

Phase / SSL Band	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Impact
III. Phase 2 <i>2030–2035</i> <i>Industrial Scaling, Standardisation & Cross-Sector Deployment</i> SSL: Emerg	Applied and industrial R&D concurrent with standardisation. Phase 2 is not purely a standardisation phase: hardware–software co-design for neuromorphic/PIM architectures proceeds (P3); sovereign EU LLM/SLM for SecOps reaches AI Act high-risk system compliance (P6); RISC-V TEE designs enter volume production via Chips JU (P1, P8); FHE performance optimisation programmes initiated (P8); ZKP-based supply chain attestation R&D (P5); Cognitive Twin prototype development (P9). <i>KPI: At least one RISC-V TEE design in volume production by 2032; sovereign EU LLM/SLM certified at AI Act high-risk level.</i>	Standardisation milestones convert pilots into binding frameworks. ETSI TC CYBER hybrid PQC profiles published by 2031 (ML-KEM per FIPS 203; ML-DSA per FIPS 204 [9,10]); CEN/CENELEC JTC 13 Simpl security MIMs ratified by 2032; 3GPP SA3/ETSI TC ITS standardised Island Mode fallback APIs published by 2031; Runtime SBOM mandate for critical infrastructure supply chains by 2031; classical-only cryptography deprecated for high-security EU government contexts by 2033; Data Visa legal instruments developed [5]. <i>KPI: CEN/CENELEC Simpl security MIMs ratified; ETSI hybrid PQC profiles published; classical-only deprecated in sovereign contexts by 2033.</i>	Continental-scale federated deployment across regulated sectors. First EU RISC-V secure element at EUCC ‘High’ by 2031–2032 [11]; hybrid PQC deployed across Industrial IoT, energy grids, and financial systems; Pan-European CTI platform operational as LLMSecOps analytics engine by 2032; UEBA baselines harmonised using STIX/TAXII and MITRE ATT&CK; multi-party privacy-preserving AI pipelines demonstrated in health and finance sectors by 2033; AI-driven graceful degradation standardised in O-RAN AI/ML framework; inter-operator east-west security interfaces operational in 2+ cross-border corridors. <i>KPI: 70%+ Common Data Spaces on Simpl-based security; APT mean dwell time reduced by 70% relative to Phase 1 baseline.</i>	EU industrial sovereignty consolidating; critical component share rising. EU Chips Act fabrication producing RISC-V TEE designs in volume; EU secure silicon reaching 30–40% market share in critical infrastructure procurement; sovereign European edge LLM/SLM in operational SecOps deployment; IPCEI-CIS outcomes translated into binding inter-operator SLA frameworks with security audit requirements; Data Visa pilots operational in Nordic–Baltic and BeNeLux corridors; competitive NaaS offers from European consortia viable. <i>KPI: EU secure silicon 30–40% critical infrastructure share; 70%+ Data Spaces on Simpl security interoperability.</i>

Phase / SSL Band	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Impact
⇒ Gate condition III → IV: R&I: First EU RISC-V TEE design certified at EUCC ‘High’ and in volume production; sovereign EU edge LLM/SLM at AI Act high-risk compliance. Policy: CEN/CENELEC Simpl security MIMs ratified; ETSI hybrid PQC profiles published; 3GPP Island Mode APIs standardised. Operational: Pan-European CTI platform operational as LLMSecOps engine; EU secure silicon exceeding 30% critical infrastructure share. Market: EU NaaS consortium offers commercially competitive in at least two cross-border corridors.				

Phase / SSL Band	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Impact	
IV. Phase 3 <i>2035–2040</i> <i>Autonomous Leadership & Frontier Innovation</i> SSL: Full Scale	Frontier research defining global state of the art. FHE reaching operational edge latency thresholds; zero-energy air interface security with ambient RF harvesting; self-evolving AI swarm protocols for autonomous Moving Target Defense; Cognitive Twin E2E orchestration for cyber-physical resilience; bio-inspired elasticity protocols standardised via ETSI ENI; battery-less node quantum-resistant authentication. EuroHPC hosts federated training infrastructure for self-improving sovereign security AI, preventing regression to non-EU foundation model dependency. <i>KPI: FHE latency within 2× AES-GCM baseline for edge analytics; EuroHPC sovereign security AI model in operational deployment.</i>	EU transitions from standard-taker to global standard-setter. EU Cloud Rulebook binding with machine-readable compliance APIs by 2037–2038; EUCC ‘High’ as default procurement requirement for all new EU ICT products; PQC-by-default mandated for all new EU ICT products by 2036; AI Act explainability requirements enforced for autonomous MTD decisions; “Schengen for Cloud” legal framework enacted, harmonising data jurisdiction across 27 Member States; ETSI/CEN/CENELEC PQC standards promoted to ISO/ITU-T as global reference implementations. <i>KPI: EU Cloud Rulebook binding by 2038; PQC-by-default mandate in force by 2036; Schengen for Cloud enacted.</i>	Fully autonomous, continent-wide sovereign security operations. EU Chips Act sovereign fabrication meeting 80%+ of European secure silicon demand by 2039; FHE operationalised for real-time edge analytics; end-to-end confidential continuum from battery-less IoT to HPC; Cognitive Twins autonomously coordinating E2E cyber-physical recovery across critical infrastructure domains; AI-driven MTD continuously shuffling network resources, IP addresses, and software diversity without human intervention; SMPC standardised for cross-border federated AI in regulated sectors (health, finance, defence, justice). <i>KPI: EU secure silicon self-sufficiency ≥80% by 2039; autonomous MTD and Cognitive Twin operational continent-wide.</i>	EU competitive parity achieved; technology standards exported globally. European operators achieving competitive scale parity with US/Chinese hyperscalers in cloud-edge NaaS; EU-designed security components dominant in European market; EUCC/EUCC adopted as global reference certification frameworks; “Schengen for Cloud” eliminating national fragmentation constraining European scale; EU cybersecurity technology export balance turning positive; SME participation economically viable without discriminatory compliance cost barriers. <i>KPI: EU NaaS competitive parity demonstrated in 3+ cross-border corridors; EUCC/EUCC referenced in ISO/ITU-T standards.</i>	

Phase / SSL Band	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Impact
$\Rightarrow$ Gate condition IV $\rightarrow$ V: R&I: FHE at operational edge latency; zero-energy authentication validated on battery-less EU hardware; AI swarm protocols stable and explainable under AI Act. Policy: EU Cloud Rulebook binding; “Schengen for Cloud” enacted; PQC-by-default mandate in force; EU standards submitted to ISO/ITU-T as reference implementations. Operational: EU secure silicon self-sufficiency $\geq 80\%$; autonomous MTD and Cognitive Twin E2E orchestration operational continent-wide. Market: EU NaaS competitive parity demonstrated; technology export balance positive.				
V. Strategic End-State Objective 2040 Vision SSL: Full A	Europe leads global security R&I agenda. EU-originated R&I outputs (PQC, TEE architectures, LLMSecOps, zero-energy security, Cognitive Twins) define the global research agenda and are adopted internationally. Alignment with SNS JU SRIA Phase 3 outcomes: 6G security primitives sovereign and globally deployed [5]. ECCC Strategic Agenda fully operationalised; European industrial cybersecurity competitiveness self-sustaining without emergency public subsidy [6]. <i>Triple accountability: SNS JU SRIA [5] $\cap$ ECCC Strategic Agenda [6] $\cap$ EU regulatory stack [1].</i>	Complete, coherent, machine-enforceable EU regulatory stack. NIS2, CRA, AI Act, DORA, Data Act, Cloud Rulebook, and “Schengen for Cloud” operating as a coherent system with machine-readable compliance APIs and no manual audit requirement. EUCS and EUCC recognised globally as reference certification frameworks. ETSI and CEN/CENELEC leading ISO/ITU-T security standardisation tracks. Compliance is automated, verifiable in real time, and does not impose discriminatory costs on SMEs or new market entrants.	Security-by-Design is the default operational baseline. Autonomous, self-healing, continent-wide security infrastructure requiring no manual intervention for routine threat response. Full confidential computing continuum from IoT sensor to HPC node. Zero critical operational dependencies on non-EU security infrastructure. Island Mode and graceful degradation are normative operating assumptions, not emergency exceptions. Bio-inspired swarm-coordinated resilience is the standard architecture for EU critical infrastructure.	Europe: sovereign, competitive, and globally influential. Europe is the global reference for sovereign, trustworthy, and sustainable computing security. EU-designed security components dominant in European market and competitive globally. SME participation in EU cybersecurity ecosystem fully economically viable. European security standards and AI-driven technologies exported as global reference implementations. Strategic autonomy measured, sustained, and no longer requiring emergency policy intervention to maintain.

Phase 1 Heterogeneity: Applied R&D vs. Fragmented Piloting

A critical methodological distinction governs the Phase 1 row across the ten executive priorities. Priorities 3 (Lightweight Cryptography and Physical-Layer Security), 6 (Cognitive SecOps), 8 (Confidential Computing and Privacy-Preserving AI Pipelines), and 9 (Resilient-by-Design Infrastructure) commence Phase 1 in an Applied R&D state because their underlying enabling technologies

— zero-energy secure authentication, EU-sovereign edge LLMs, RISC-V TEE hardware, and Cognitive Twin orchestration respectively — are not deployable at operational scale as of 2025–2026. The remaining six priorities commence Phase 1 in a Fragmented Piloting state because near-TRL solutions exist and regulatory pull from CRA, NIS2, DORA, and the AI Act is sufficient to drive pilot deployments in the near term.

This distinction is not cosmetic. Treating all ten priorities as uniform pilots in Phase 1 would constitute a material overclaim of maturity to both IEEE peer reviewers and European Commission technical evaluators, and would produce internally inconsistent recommendations by implying that technologies currently at TRL 3–4 can be mandated in operational deployments by 2026. When instantiating per-priority progression matrices, the R&I cell for Priorities 3, 6, 8, and 9 in Phase 1 must state Applied R&D explicitly; it must not claim pilot or deployment activity.

Usage as a Consistency Reference

The Master Capability Transition Framework serves three functions in the broader roadmap document. First, it provides the structural skeleton within which each per-priority progression matrix in Section 7 is located: every cell in a per-priority matrix should be traceable to the corresponding intersection of phase row and analytical dimension in Table 1. Second, it provides a cross-priority consistency check: if two per-priority matrices claim the same gate condition (e.g., EU Chips Act fabrication capacity, which gates both Priority 1 Phase 2 and Priority 8 Phase 2), that shared dependency should be explicitly visible in the master framework and flagged in the executive summary as a single-point-of-failure risk for the roadmap. Third, the KPI anchors embedded in each phase cell provide a minimum measurable outcome that must be demonstrated before the corresponding gate condition can be declared satisfied; these KPIs are derived from the per-priority matrices and are reproduced here as cross-reference anchors only.

Security Framework with Separated Security Verticals

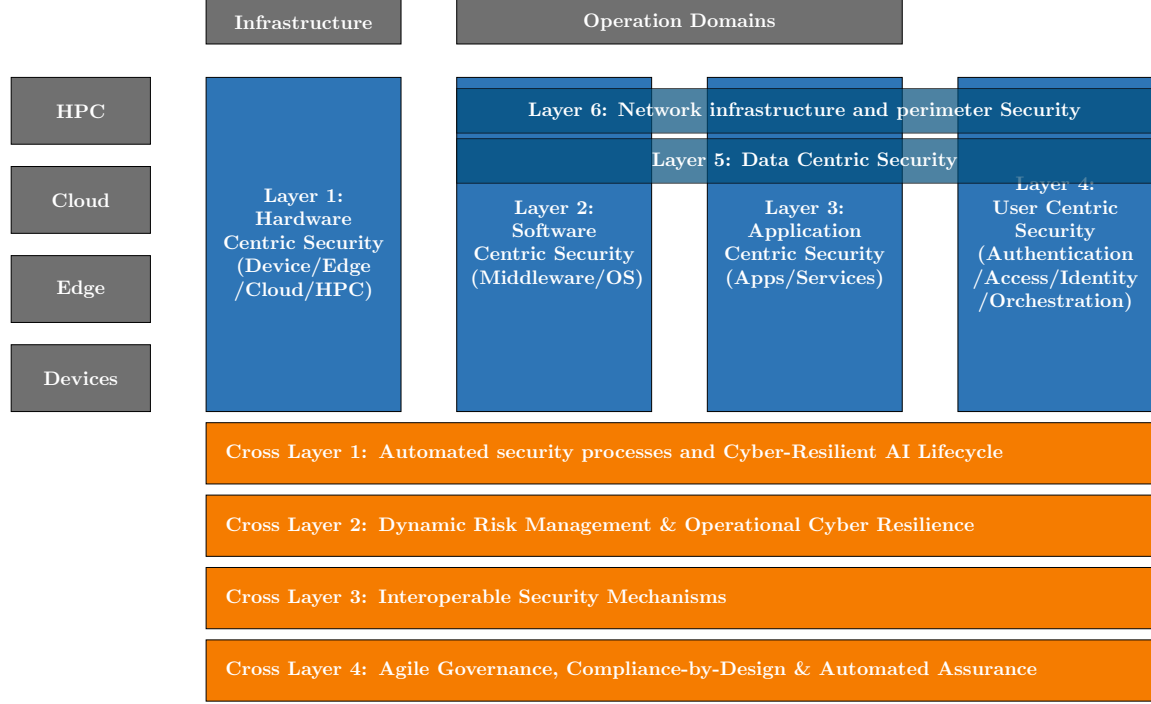


Figure 1: Security Framework with separated security verticals in dedicated boxes.

5 Methodology for Cybersecurity Prioritization in the Computing Continuum - To Process

The roadmap is developed through a structured five-stage analytical pipeline (Figure 1). Each stage has a clearly defined input, transformation logic, and output, ensuring that the final strategic recommendations are fully traceable back to the architectural framework and threat landscape.

The architectural framework presented in Figure 1 serves as the formal reference model of this roadmap. It defines the logical structure of cybersecurity capabilities across the Cloud–Edge–IoT continuum and provides the structured methodological basis from which candidate capabilities are derived for prioritization.

1. **Stage 1 – Capability Identification:** Using the architectural framework as the structured reference model, this stage systematically identifies and enumerates the cybersecurity capabilities required across the Cloud–Edge–IoT continuum. The framework organizes these capabilities into six vertical Security Layers and four transversal Cross-Layers, ensuring comprehensive coverage of technical, operational, and governance functions. Each sub-item within this

framework constitutes a distinct *candidate capability* that enters the prioritization pipeline.

2. **Stage 2 – 7-Dimensional Capability Profile:** Each Executive Priority is analyzed in depth through a seven-dimension template, producing a complete, actionable R&I profile: where Europe stands today, what must be built, why it is urgent, how to get there, what could go wrong, and how the system survives failure (Section 5.3).
3. **Stage 3 – Progression Matrix and Cross-Priority Phase Synthesis:** Each priority’s 7-dimensional profile is condensed into a three-phase progression matrix tracking two objective metrics: Operational Deployment Phase (ODP) and Strategic Sovereignty Level (SSL). The matrices are then read *across* all priorities per phase to identify shared enablers, cross-cutting risks, and aggregate strategic postures for each time horizon (Section 5.4).
4. **Stage 4 – 4-Filter Prioritization:** Each candidate priority is assessed against four filters. Priorities satisfying the highest number of filters, and in particular satisfying Filters 1 or 2 as mandatory conditions, are elevated to *Executive Priorities*. The output of this stage is an ordered set of priorities defining the Phase 1 critical investment path (Section 5.2).
5. **Stage 5 – Executive Summary Synthesis:** The per-phase strategic postures produced in Stage 4, combined with the cross-phase execution principles, are synthesized into phase-specific strategic recommendations and measurable success metrics for presentation to European Commission evaluators, funding bodies, and Member State national cybersecurity agencies (Section 5.5).

5.1 Stage 1: Candidate Capability Identification

The architectural framework presented in Figure 1 serves a dual purpose: it is both a reference architecture for the secure European computing continuum *and* the structured source from which candidate capabilities are derived for prioritization.

Every named sub-item within the six Security Layers (Layers 1 through 6) and the four Transversal Cross-Layers (CL1 through CL4) constitutes a distinct *candidate capability* with an independent security function, maturity trajectory, and set of R&I requirements. The two structural dimensions of the framework impose an important ordering constraint on the pipeline:

- **Vertical dependency (Layer hierarchy):** Lower layers provide cryptographic and physical anchors upon which higher layers depend. A capability in Layer 4 (Orchestration) cannot provide meaningful security guarantees unless the capabilities in Layers 1 and 2 are already trustworthy. This architectural dependency directly informs Filter 2 of the prioritization stage.
- **Horizontal reach (Cross-Layer scope):** Cross-Layer capabilities are not confined to a single technical layer; they apply governance, automation, and interoperability functions *across* the full stack. Their prioritization reflects their role as force-multipliers for multiple vertical capabilities simultaneously, which is directly assessed by Filters 3 and 4.

The complete enumeration of candidate capabilities is presented in Section 6. Every candidate that passes through Stage 2 retains its Layer or Cross-Layer identifier, ensuring full architectural traceability throughout the roadmap.

5.2 Stage 2: The 4-Filter Prioritization Framework

To provide actionable guidance for Horizon Europe and European Cybersecurity Competence Centre (ECCC) investment decisions, each candidate capability is assessed against a structured four-filter logic. The objective is to identify the *critical-path* capabilities requiring immediate resource allocation during Phase 1 (0–5 years).

The four filters are not applied as an all-or-nothing gate; rather, each filter that a capability satisfies contributes a *positive signal*. Two classification rules apply:

- A capability satisfying **Filter 1 or Filter 2** is unconditionally elevated to an Executive Priority, because regulatory non-compliance and architectural blocking both represent immediate, non-deferrable risks.
- A capability satisfying **Filters 3 and/or 4** but neither Filter 1 nor 2 is scheduled as a Phase 2 or Phase 3 priority, with its rank determined by the number of filters satisfied.

1. Filter 1: Regulatory Urgency (The “Must-Do” Filter)

- **Question:** Is this capability required to meet imminent European regulatory obligations (NIS2 [1], Cyber Resilience Act [2], AI Act [3])?
- **Logic:** Capabilities whose absence by 2026–2027 results in market exclusion, systemic non-compliance, or critical infrastructure vulnerability are unconditionally elevated to Phase 1 priority. The enforcement timeline of the relevant regulation determines urgency: CRA product requirements, NIS2 incident reporting obligations, and AI Act high-risk system conditions each impose distinct, legally binding deadlines.
- **Output signal:** *Mandatory – Phase 1 (non-deferrable).*

2. Filter 2: Foundational Dependency (The “Blocker” Filter)

- **Question:** Do other security capabilities depend on this capability as an architectural prerequisite?
- **Logic:** Capabilities lower in the vertical stack (e.g., Layer 1 Hardware Roots of Trust) must be established before higher-layer capabilities (e.g., Layer 4 Zero-Trust Orchestration or Layer 5 Confidential Computing) can provide meaningful guarantees. A blocked dependency means that funding higher-layer capabilities before the lower-layer anchor exists wastes investment. The architectural layer numbering (Section 5.1) provides the formal precedence rule.
- **Output signal:** *Mandatory – Phase 1 (architectural prerequisite).*

3. Filter 3: Strategic Sovereignty Risk (The “Extinction” Filter)

- **Question:** Does the absence of this capability create a strategic dependency, systemic resilience gap, or technology cliff for European digital sovereignty?
- **Logic:** This filter identifies capabilities where (a) Europe currently holds 100% dependency on non-EU vendors, leaving critical infrastructure exposed to extra-territorial access and geopolitical leverage, or (b) a foreseeable adversarial or technological breakthrough would render existing European defenses obsolete before countermeasures can be deployed. The “sovereignty cliff” designation applies when either condition is true and no credible EU-controlled fallback exists.

- **Output signal:** *High priority – Phase 1 or Phase 2 depending on co-occurrence with Filters 1/2.*

4. Filter 4: Market Feasibility & Ecosystem Alignment (The “Adoption” Filter)

- **Question:** Can this capability realistically be adopted by European industry—including SMEs—and does it align with and amplify existing EU investments (e.g., Simpl middleware [7], EU Chips Act, Digital Europe Programme)?
- **Logic:** Priority is given to capabilities that act as force-multipliers for existing European infrastructure, reducing duplication and ensuring economic viability for the full ecosystem. A capability with strong market feasibility but weak filter 1/2/3 scores is deferred to a phase where adoption infrastructure is mature.
- **Output signal:** *Amplifying factor – influences phase placement and implementation sequencing.*

Stage 2 Output: The multi-filter assessment produces an ordered set of *Executive Priorities* — the capabilities that define the Phase 1 critical investment path — plus a ranked backlog of Phase 2 and Phase 3 capabilities. Each Executive Priority carries a filter-satisfaction record that directly justifies its rank in the 7-dimensional profiles of Stage 3. The current roadmap identifies ten Executive Priorities (detailed in Section 7), with Priorities 1–3 satisfying the highest number of filters and constituting the absolute critical path.

5.3 Stage 3: The 7-Dimensional Capability Profile

Each Executive Priority produced by Stage 2 is translated into a complete, evaluator-grade R&I profile using a seven-dimension analytical template. This template ensures that every capability is characterized not only from a technical perspective but also in terms of its regulatory alignment, investment pathway, failure modes, and contribution to European strategic autonomy. The template is applied identically to every priority, enabling consistent cross-priority comparison.

1. Dimension 1: Baseline & Target Objective

Question: *Where does Europe stand today, and what does success look like by 2040?*

Purpose: Grounds the priority in reality by capturing the current Operational Deployment Phase (ODP), Strategic Sovereignty Level (SSL), industrial capacity, and barriers to adoption. The target defines measurable end-states (target ODP, target SSL, quantified EU market share) that serve as the basis for funding project calls and Phase 4 synthesis.

2. Dimension 2: Primary Drivers

Question: *Why must Europe build this capability now?*

Purpose: Articulates the external pressures—both adversarial and regulatory—that make this priority urgent and non-deferrable.

- **Threat Drivers:** Adversarial capabilities or attack vectors that are already active or imminently scalable.
- **Regulatory Drivers:** Specific EU legal obligations with enforcement timelines, citing the precise directive and relevant provisions.

3. Dimension 3: High-Impact Activities

Question: Which specific technical interventions yield the greatest strategic return on investment?

Purpose: Identifies 3–5 concrete, actionable R&I activities within the capability area, each formulated as a specific, measurable initiative rather than a general goal. Activities are tagged with their target phase (Phase 1, 2, or 3) to ensure the progression matrix in Stage 4 is anchored in concrete work.

4. Dimension 4: Catalysts & Enablers

Question: What specific EU funding, standards, and ecosystem support are required to accelerate from baseline to target?

Purpose: Identifies the concrete pull mechanisms and investment vehicles needed to drive market adoption.

- **R&I & Funding Alignment:** Horizon Europe clusters, Digital Europe Programme calls, and ECCC Strategic Agenda work packages [6].
- **Regulatory & Standardization:** Required CEN/CENELEC, ETSI, or ISO standards, distinguishing between existing standards to be referenced and new standards that must be developed.
- **Ecosystem Enablers:** Pilot infrastructure (e.g., federated Cyber Ranges), industrial partnerships (e.g., Gaia-X, IPCEI-CIS), and SME capacity-building schemes.

5. Dimension 5: Disruptors & Systemic Risks

Question: What structural vulnerabilities, market failures, or adversarial breakthroughs could derail the capability trajectory?

Purpose: Acknowledges that R&I trajectories are non-linear by identifying the specific risks that could render planned deployments obsolete or unachievable. Three risk categories are assessed for each priority:

- **Technical Cliffs:** Moments when a defensive capability is rendered obsolete by an adversarial or technological breakthrough before the EU countermeasure is fully deployed (e.g., a cryptanalytic advance against a standardized algorithm before migration is complete; a side-channel attack class that bypasses current hardware enclave implementations).
- **Supply Chain & Market Risks:** Fabrication bottlenecks, vendor resistance to open standards, monopolization of critical components by non-EU providers, or SME exclusion from certification pathways.
- **Geopolitical & Regulatory Risks:** Export controls on critical components, extra-territorial jurisdiction conflicts, or fragmented national implementation undermining EU-wide interoperability.

6. Dimension 6: Failsafe Mechanisms & Resilience

Question: If this capability is compromised or delayed, how does the system survive and gracefully degrade?

Purpose: The technical core of “Cyber-Resilience” as applied to the roadmap itself. It defines engineering fallbacks for each priority’s primary mechanism. Four categories of failsafe must be addressed:

- **Hybrid Transitional Schemes:** Parallel deployment of the target technology alongside the incumbent (e.g., classical + PQC cryptography; proprietary TPM + RISC-V secure element) during the migration window.
- **Graceful Degradation:** The minimum viable operational mode the infrastructure can sustain if the primary capability is unavailable (e.g., cached credential operation during cloud disconnection; localized physical-layer authentication if cryptographic processing fails).
- **Autonomous Isolation Protocols:** Automated quarantine or revocation mechanisms that activate when a node or component fails attestation, preventing compromise propagation without human intervention.
- **Implementation Diversity:** Deliberate multi-vendor or multi-algorithm diversity to avoid monoculture vulnerabilities (e.g., deploying two independent TEE architectures; supporting multiple NIST-standardized PQC algorithm families).

7. Dimension 7: Transversal Meta-Layer (Policy Alignment)

Question: How does mastering this capability advance Europe’s overarching goals of sovereignty, sustainability, and societal trust?

Purpose: Elevates the technical capability profile to the political and societal level, connecting it to the EU’s strategic autonomy agenda.

- **Sovereignty & Autonomy:** Quantified reduction in extra-territorial technology dependencies; contribution to SSL progression.
- **Trust & Ethics:** Alignment with GDPR, Privacy-by-Design, algorithmic transparency, and the AI Act’s trustworthiness requirements.
- **Sustainable Security (Green IT):** Energy efficiency of cryptographic or monitoring operations; e-waste reduction through crypto-agility (eliminating hardware replacement); alignment with the EU Green Deal.

Stage 3 Output: A complete, 7-dimension capability profile for each Executive Priority. The Baseline (Dimension 1) and High-Impact Activities (Dimension 3) directly populate the *Operational State* cells of the Stage 4 Progression Matrix. The Critical Enablers (Dimension 4) populate the matrix’s *Critical Enabler* row. Disruptors (Dimension 5) and Failsafes (Dimension 6) inform the resilience provisions in the Phase Synthesis and Executive Summary.

5.4 Stage 4: Progression Matrix and Cross-Priority Phase Synthesis

4a. The Per-Priority Progression Matrix

Rather than relying on subjective Technology Readiness Levels (TRL) for a highly fragmented ecosystem, each capability profile is condensed into a three-phase progression matrix using four objective, market-observable indicators. The matrix columns correspond to the three strategic time horizons; each row tracks a distinct dimension of maturity:

- **Operational Deployment Phase (ODP):** Tracks the market reality and integration status of the capability across the EU ecosystem. The ODP scale progresses through four stages: *Applied R&D* (laboratory/prototype phase) → *Fragmented Piloting* (isolated national or sectoral pilots) → *Standardized Integration* (harmonized deployment across multiple providers

and Member States) → *Ubiquitous Operation* (Security-by-Design default, manual configuration obsolete). A capability’s *starting* ODP in Phase 1 is set by its Dimension 1 Baseline; the *target* ODP in Phase 3 is set by its Dimension 1 Target.

- **Strategic Sovereignty Level (SSL):** Tracks European strategic autonomy and dependency on non-EU technology vendors. The SSL scale progresses through three stages: *High Foreign Dependency* (critical function relies on non-EU components with no credible EU alternative) → *Emerging EU Ecosystem* (EU alternatives exist at pilot or early production scale; hybrid EU/non-EU deployments) → *Full Sovereign Autonomy* (EU-designed components dominate the market; no critical extra-territorial dependency). SSL progression is gated by the concrete investments and fabrication milestones identified in Dimension 4.
- **Operational State:** A concise 2–3 sentence snapshot characterizing the practical reality “on the ground” during each phase: which deployments are active, what remains manual or incomplete, and what the dominant risk exposure is at that point in time.
- **Critical Enabler:** The single most important regulation, standard, or European R&I breakthrough that must be achieved to *unlock the transition* to the next phase. This is not a list of desirable inputs but the specific gate condition without which Phase transition cannot occur. It is derived directly from Dimension 4 (Catalysts & Enablers) and Dimension 5 (Disruptors) of the capability profile.

The three time horizons are defined as follows:

- **Phase 1: Foundation & Compliance (0–5 Years):** Focus on state-of-practice deployments, immediate regulatory compliance (NIS2, CRA enforcement), and bridging the most critical capability gaps. The ODP and SSL positions at the end of Phase 1 define the baseline for Phase 2 investment planning.
- **Phase 2: Federation & Standardization (5–10 Years):** Focus on industrializing and federating Phase 1 pilots, establishing harmonized governance frameworks, and managing critical technology transitions (e.g., PQC migration, RISC-V silicon scale-up). Cross-border interoperability and pan-European certification systems become the primary integration challenge.
- **Phase 3: Autonomy & Leadership (10–15 Years):** Focus on achieving full sovereign autonomy, deploying autonomous Security-by-Design architectures, and positioning European frameworks as global reference implementations. Residual Phase 2 transitions are completed; next-generation threat vectors (post-quantum, neuromorphic, space-based edge) are addressed.

4b. Cross-Priority Phase Synthesis

The per-priority matrices are individual capability views. Before producing the Executive Summary, the matrices must be read *across* all priorities within each phase to generate a *phase-level strategic posture assessment*. This cross-priority synthesis follows three analytical steps:

1. **Shared Critical Enabler Identification:** The *Critical Enabler* row is read across all priority matrices for a given phase. Enablers that appear for multiple priorities in the same phase

(e.g., “CRA enforcement” appearing as the Phase 1 enabler for both Hardware Roots of Trust and Supply Chain Verification) are designated as *cross-cutting investments* — regulatory or funding actions with disproportionate leverage across the portfolio.

2. **ODP/SSL Posture Aggregation:** The ODP and SSL positions across all priorities at the end of a given phase are reviewed collectively to identify: (a) *lagging capabilities* — where ODP or SSL progression is significantly behind the portfolio median, indicating that a shared bottleneck (fabrication, standardization, political will) is constraining multiple priorities simultaneously; (b) *interdependency risks* — where a Phase 2 capability’s ODP starting point depends on a Phase 1 capability reaching a specific SSL level that is currently projected to be insufficient.
3. **Phase Gate Assessment:** For each phase, the synthesis produces a single *phase gate assessment*: a structured statement of (a) the cross-cutting enabling actions that unlock the most capability transitions simultaneously, (b) the primary shared risk that could stall the portfolio, and (c) the aggregate ODP/SSL trajectory European infrastructure is on. This phase gate assessment is the direct input to Stage 5.

Stage 4 Output: Three phase gate assessments (one per time horizon) and an identified set of cross-cutting investments, both of which feed directly into the Executive Summary recommendations.

5.5 Stage 5: Executive Summary Synthesis

The Executive Summary (Section 9) is not intended as a standalone narrative but the direct output of the Stage 4 phase gate assessments combined with the cross-phase execution principles.

Its structure maps one-to-one onto the pipeline outputs:

- **Strategic Narrative:** Synthesizes the overall ODP/SSL trajectory across all priorities into a coherent description of the journey from today’s fragmented, foreign-dependent posture to full sovereign autonomy.
- **Per-Phase Recommendations:** Each phase’s cross-cutting investments (from Phase Synthesis, Step 1) are restated as actionable policy and funding directives for European Commission evaluators and Member State agencies.
- **Success Metrics:** The measurable targets stated in each priority’s Dimension 1 (Target Objective) and the Phase 1/2/3 ODP/SSL end-states from the matrices are aggregated into a unified set of per-phase success criteria, enabling objective evaluation by ECCC annual reviews.
- **Risk Register:** The primary shared risk identified in each Phase Gate Assessment is surfaced as a named strategic risk requiring cross-portfolio mitigation, not merely a per-priority concern.

5.6 The Anatomy of a Roadmap Priority: A Hybrid 7-Dimensional Analysis Template

To ensure the roadmap is actionable, realistic, and aligned with European strategic autonomy, each priority is analyzed through a hybrid 7-dimensional profiling framework that combines methodolog-

Table 2: Priority profiling dimensions and applicability

Dimension	Type	Purpose
Baseline & Target Objective	Mandatory	Define maturity trajectory and target state
Primary Drivers	Mandatory	Establish urgency and justification
High Impact Activities	Mandatory	Identify key interventions
Catalysts & Enablers	Mandatory	Define EU support mechanisms
Disruptors & Risks	Adaptive	Identify systemic and adversarial threats
Failsafe Mechanisms & Resilience	Adaptive	Ensure resilience under failure
Transversal Meta-Layer	Adaptive	Align with EU policy and societal goals

ical consistency with contextual flexibility. The template is applied consistently across priorities, with the first four dimensions treated as mandatory and the remaining dimensions applied where relevant based on risk exposure, system criticality, and policy impact.

Priority Profiling Framework: Mandatory and Adaptive Dimensions

To ensure both analytical rigor and flexibility across heterogeneous priorities, the 7-dimensional profiling framework is structured into:

- **Mandatory Core Dimensions:** Applied to every priority to ensure comparability, traceability, and alignment with EU strategic objectives.
- **Adaptive Contextual Dimensions:** Applied selectively depending on the nature, maturity, and risk profile of each priority.

This hybrid approach ensures that all priorities are evaluated consistently, while allowing deeper analysis where complexity, uncertainty, or systemic risk requires it.

Dimension Applicability Logic

Not all priorities require equal depth across all seven dimensions. Therefore:

- The **first four dimensions are mandatory** and must be defined for every priority.
- The remaining three dimensions are **conditionally applied** based on:
 - System criticality
 - Exposure to systemic or adversarial risk
 - Policy and societal relevance

This ensures proportional analysis effort while maintaining methodological consistency.

1. **Baseline & Target Objective (The Trajectory) - Q: Where do we stand today, and what exactly does success look like?**

Description: Establishes the starting point of the priority today and defines the ultimate maturity state required by the end of the 15-year roadmap.

Evaluation Output: Defines the maturity gap and informs phase transition readiness.

- **Baseline (Where we are):** Grounds the roadmap in reality. It captures the state of the art, current Operational Deployment Phase (ODP), Strategic Sovereignty Level (SSL), industrial capacity, reliance on foreign vendors, and the technical or economic barriers preventing immediate adoption.
- **Target (Where we need to be):** Sets the baseline for EU-funded projects, defining the target ODP and SSL that must be achieved by the end of the roadmap.

2. **Primary Drivers (The “Push” Factors) - Q: Why must we prioritize building this capability right now? SQ: Why are we doing this?**

Description: Outlines the external pressures—both from the evolving threat landscape and mandatory EU regulations—that make this technological evolution urgent and unavoidable.

Interpretation: Determines urgency and influences prioritization ranking.

- **Threat Drivers:** The rapid advancement of adversarial capabilities (e.g., offensive AI, quantum decryption, supply chain targeting).
- **Regulatory Drivers:** Imminent EU compliance mandates that require technical operationalization (e.g., NIS2, Cyber Resilience Act, AI Act).

3. **High Impact Activities - Q: Which specific actions provide the greatest leap in European sovereignty and security? SQ: Which should be the key initiatives?**

Description: Highlights the specific, prioritized technical interventions within this broader capability group that will yield the highest strategic return on investment.

Output: A shortlist of actionable R&I and deployment interventions.

4. **Catalysts & Enablers (The “Pull” Factors & EU Support) - Q: What specific EU funding, standards, and ecosystem support do we need? SQ: How do we get there & accelerate?**

Description: Identifies the concrete mechanisms, funding vehicles, and regulatory standardizations required from the EU to accelerate market adoption and push the industry from the baseline to the target state.

Role in Methodology: Bridges priorities with EU instruments, ensuring implementation feasibility.

- **R&I & Funding Alignment:** Required Horizon Europe funding topics and targeted innovation instruments. Investment priorities must be explicitly synchronized with the European Cybersecurity Competence Centre (ECCC) Strategic Agenda [6] to ensure alignment with pan-European goals (e.g., PQC transition infrastructure, AI-driven security tools).
- **Regulatory & Standardization:** Alignment with standard-setting bodies (CEN/CENELEC) and certification frameworks (e.g., EUCS, EUCC).

- **Ecosystem:** Support for ecosystem building, industrial partnerships, and capacity-building initiatives (e.g., federated Cyber Ranges) to upskill the workforce and ensure advanced security solutions are economically viable for SMEs.
5. **Disruptors & Systemic Risks (The Roadblocks) - Q: What structural vulnerabilities, market failures, or adversary breakthroughs could derail us? SQ: What can break the path?**
- Description:* Acknowledges that innovation is not linear by identifying severe technical and architectural risks that could actively derail deployment. It places primary emphasis on “Technical Cliffs”—moments where defensive capabilities are rendered obsolete by adversarial breakthroughs before countermeasures are fully deployed.
- Trigger Condition:** Applied when the priority is exposed to technological discontinuities or adversarial asymmetry.
- **Cryptographic Obsolescence & Migration Bottlenecks (The Primary Technical Risk):**
 - *The “Q-Day” Cliff:* The risk that Cryptographically Relevant Quantum Computers (CRQC) become operational before the EU completes its migration to Post-Quantum Cryptography (PQC). This creates a “Harvest Now, Decrypt Later” vulnerability.
 - *Migration Complexity:* The systemic failure to inventory and replace deep-rooted Public Key Infrastructure (PKI) in legacy OT/IoT environments.
 - **Adversarial Breakthroughs:** The risk of threat actors weaponizing emerging technologies (e.g., AI-driven exploit generation, polymorphic malware) faster than defensive R&I can mitigate them.
 - **Contextual Risks (Non-Technical / Optional):** Geopolitics & supply chain dependencies, market & technology risks (monopolization, fragmented standards).
6. **Failsafe Mechanisms & Resilience (The Safety Nets) - Q: If this technology is compromised, how does the system survive and gracefully degrade? SQ: How do we prevent failure?**
- Description:* The technical core of “Cyber-Resilience.” It defines the engineering fallbacks required if a primary capability fails, is breached, or is disrupted by the systemic risks identified above.
- Purpose:** Ensures continuity of operations under failure or compromise scenarios.
- What it entails:** Designing for degraded modes of operation, architectural redundancies, autonomous isolation protocols, and implementation diversity to avoid vulnerable technological monocultures.
7. **Transversal Meta-Layer (The Policy Alignment) - Q: How does mastering this capability support Europe’s goals of sovereignty, sustainability, and trust? SQ: The broader policy context?**
- Description:* Elevates the technical capability back to the overarching political and societal goals of the European Union.
- Strategic Role:** Aligns technical priorities with EU-wide policy narratives and long-term societal impact.

Table 3: Mapping from 7-Dimensional Profile to Progression Matrix

Profile Dimension	Matrix Component
Baseline & Target Objective	ODP and SSL starting/target levels
High Impact Activities	Operational State evolution
Catalysts & Enablers	Critical Enabler
Disruptors & Risks	Phase risk considerations
Failsafe Mechanisms	Resilience assumptions in synthesis

- **Sovereignty & Autonomy:** Strengthening European strategic independence, reducing reliance on non-EU vendors.
- **Trust & Ethics:** Fostering deep societal trust by guaranteeing data privacy (Privacy-by-Design), algorithmic fairness, and transparency.
- **Sustainable Security (Green IT):** Minimizing the environmental impact of cyber defense (e.g., energy-efficient cryptographic algorithms, crypto-agility to reduce e-waste).

5.7 Phase Transition Mechanism - Progression Matrix: Sequence of Events (X-Axis)

The roadmap is structured across multiple time phases (e.g., short-term, mid-term, long-term). Each phase represents a column in the prioritization matrix.

Transitions between phases are governed by the evolution of capabilities and priorities across these columns.

Instead of a strictly chronological timeline, the roadmap maps the sequence of critical events required to unlock maturity. This ensures that foundational capabilities are built before advanced autonomous systems.

- **Phase 1: Foundation & Compliance (Immediate Implementation):** Focuses on state-of-practice deployments, bridging immediate capability gaps, and addressing urgent regulatory obligations (NIS2, CRA).
- **Phase 2: Federation & Standardization (Scaling & Harmonization):** Focuses on deploying emerging technologies, establishing harmonized governance frameworks, and managing critical transition periods.
- **Phase 3: Autonomy & Leadership (Sovereign Innovation):** Focuses on highly speculative but structured foresight, including defense against quantum-capable adversaries and the achievement of fully sovereign, autonomous cloud-edge ecosystems.

For each phase:

- Input: Set of prioritized capabilities from the previous phase
- Processing: Application of the 4-dimensional filtering logic
- Output: Updated set of priorities for the next phase

A transition occurs when:

- A capability increases in maturity or feasibility
- Dependencies are resolved
- Regulatory pressure increases
- Strategic importance shifts

Each transition produces a **phase summary**, capturing:

- Newly introduced priorities
- Evolved priorities
- Deferred or dropped priorities

Matrix Cell Content Definition (Measuring Progression)

Rather than relying on subjective Technology Readiness Levels (TRL) or Manufacturing Readiness Levels (MRL) for a highly fragmented ecosystem, each cell in the progression matrix tracks maturity using objective strategic market and policy indicators:

- **Operational Deployment Phase (ODP):** Tracks the market reality and integration status (e.g., *Applied R&D* → *Fragmented Piloting* → *Standardized Integration* → *Ubiquitous Operation*).
- **Strategic Sovereignty Level (SSL):** Tracks European strategic autonomy and reliance on foreign vendors (e.g., *High Foreign Dependency* → *Emerging EU Ecosystem* → *Full Sovereign Autonomy*).
- **Operational State:** A concise snapshot characterizing the reality on the ground during this phase.
- **Critical Enabler:** The single most critical regulation, standard, or European R&I breakthrough unlocking the transition to the next phase.

5.8 From Priorities to Executive Summary: Final Strategic Synthesis

After priorities are profiled and their cross-phase progression is captured in the transition matrix, a final synthesis step is applied to convert detailed technical analysis into executive-level strategic guidance.

The synthesis is non-normative and evidence-driven, relying exclusively on outputs generated in previous stages rather than introducing new assumptions.

This synthesis does not re-score priorities. Instead, it aggregates the strongest signals emerging from the previous methodological steps and translates them into decision-oriented messages for policymakers, funders, and industrial stakeholders.

Inputs to the Synthesis Step

The final synthesis reads and consolidates the following inputs:

- The selected **Executive Priorities** emerging from the 4-filter logic
- The **mandatory and adaptive dimensions** captured in each priority profile
- The **cross-phase evolution** recorded in the progression matrices, including ODP, SSL, operational state, and critical enablers

Synthesis Logic

This step defines the **aggregation logic** of the methodology, transforming individual priority analyses into system-level strategic insights. The synthesis step identifies:

- **Common patterns** across priorities (e.g., repeated dependencies on hardware trust, PQC migration, interoperability, or compliance automation)
- **Cross-cutting bottlenecks** that appear in multiple priorities and therefore require coordinated EU action
- **Strategic inflection points** where delays in one capability would cascade into multiple downstream priorities
- **Investment signals** indicating where Horizon Europe, Digital Europe, ECCC, or standardization efforts should be concentrated

Output of the Synthesis Step

The output of this step is an executive summary that distills the roadmap into:

- A concise set of **critical strategic messages**
- A **prioritized investment narrative** for Phase 1 and beyond
- A **policy-aligned explanation** of why these priorities matter for sovereignty, resilience, and trust

This ensures that the executive summary is fully traceable to the analytical pipeline rather than functioning as a standalone narrative.

5.9 Executive Summary as Aggregated Output

The executive summary represents the final synthesis of the methodology and the output of the strategic synthesis step. It is not an independent narrative, but a structured aggregation of:

- High-priority capabilities (from filtering)
- Cross-phase evolution trends (from transitions)
- Strategic gaps and investment needs

It provides:

- A consolidated view of critical priorities
- Recommendations for R&I funding and policy alignment
- Identification of strategic dependencies and risks

This ensures full traceability from raw capabilities to strategic recommendations.

6 Capabilities (Architectural Framework)

As it is illustrated in Figure 1, the roadmap is structured across a cohesive architectural framework that spans four continuum environments (HPC, Cloud, Edge, Devices). The capabilities are organized into Six Security Layers (The Vertical Stack) and Four Transversal Cross-Layers (The Horizontal Enablers).

Part A: The Security Layers (Vertical Stack)

These layers define the specific technical security controls implemented across the computing infrastructure.

Layer 1: Hardware-Centric Security

1.1 Hardware Roots of Trust: Cryptographically verifiable supply chain trust (e.g., TPM attestation, secure enclaves) ensures physical device integrity. Success depends on driving European open-hardware designs (e.g., RISC-V) toward **EUCC 'High' assurance level certification** [11] to mathematically guarantee resistance against state-level physical tampering and side-channel attacks at the vulnerable far-edge. **1.2 Physical-Layer (PHY) Threat Detection & Fingerprinting:** Exploiting intrinsic wireless channel characteristics, such as Channel State Information (CSI), Carrier Frequency Offset (CFO), and Received Signal Strength Indicator (RSSI). That is to establish real-time, keyless device authentication and anomaly detection at the lowest layer of the stack. This provides a hardware-anchored security primitive that is resilient even when upper-layer cryptographic processing is compromised or infeasible.

Layer 2: Software-Centric Security (Middleware/OS)

2.1 OS & Middleware Hardening: Securing hypervisors, container runtimes, and distributed operating systems is critical for isolation. This involves hardening KVM/QEMU and supporting sovereign, lightweight Linux distributions optimized for edge resource constraints, ensuring a reduced attack surface for the cognitive continuum.

2.2 Secure OTA Patch & Update Management: Edge nodes deployed in the field—smart grids, O-RAN radios, industrial gateways—require cryptographically verified, failure-resilient over-the-air (OTA) update pipelines. A compromised or failed update mechanism can render entire fleets of field-deployed nodes permanently vulnerable, making robust OTA management a critical software-layer capability.

Layer 3: Application-Centric Security (Apps/Services)

3.1 Workload Security: Securing containerized applications, microservices, and serverless functions executing dynamically across multi-provider nodes. This is achieved through hardened service meshes and sidecar proxy architectures that provide mutual TLS and granular traffic inspection without impacting performance.

3.2 In-Band Network Telemetry (INT) & Deep Observability: Embedding “smart telemetry” directly into the data packets traversing containerized workloads and service meshes, providing granular, real-time visibility into network behaviour and performance without requiring

out-of-band monitoring infrastructure. This feeds high-fidelity signals to security analytics platforms.

3.3 API Security Gateway & Policy Enforcement: As containerized AI workloads increasingly communicate through REST/gRPC APIs, a dedicated API security gateway provides a unified enforcement point for authentication tokens, rate limiting, schema validation, and payload inspection—preventing injection attacks and data exfiltration at the application boundary.

Layer 4: User & Orchestration-Centric Security

4.1 Zero-Trust Orchestration: Seamless policy enforcement across a multi-provider computing continuum requires robust Policy Enforcement Point (PEP) and Policy Decision Point (PDP) architectures. This enables continuous authorization and dynamic access controls that adapt to the context of the request.

4.2 Federated Identity & Access Management (IAM): Cross-domain identity federation and verifiable credentials for machines, users, and automated workloads. Implementing **Self-Sovereign Identity (SSI)** principles allows for cryptographically secure, decentralized authentication across heterogeneous cloud-edge domains.

4.3 Privileged Access Management (PAM) & Just-In-Time (JIT) Access: Lateral movement by attackers frequently exploits over-provisioned standing privileges—the precise vector behind major Advanced Persistent Threat (APT) campaigns. PAM and JIT access controls ensure that elevated privileges are granted only for defined, time-bounded sessions with full auditability, minimizing the blast radius of any credential compromise.

Layer 5: Data-Centric Security

5.1 End-to-end Confidential Computing: Data-in-use security through Trusted Execution Environments (TEEs) spanning the entire Cloud-to-IoT data path. Leveraging TEE abstraction layers and remote attestation ensures that sensitive European data remains encrypted even during processing in non-trusted environments.

5.2 Post-Quantum & Cryptographic Agility: PQC standardization and hybrid crypto deployment are necessary to secure long-lived data. Priority is placed on **PQC migration for resource-constrained devices**, ensuring that edge infrastructure is resilient against “Harvest Now, Decrypt Later” quantum threats.

5.3 Signaling (Data) Security: Moving beyond best-effort packet forwarding by embedding trust directly into the network signaling architecture. This ensures the integrity and authenticity of control-plane messages, preventing hijacking of routing decisions and network configuration across the distributed continuum.

5.4 Distributed Reputation & Trust Networking: Implementing a ubiquitous, real-time reputation system built on Distributed Ledger Technology (DLT) and smart contracts. This enables autonomous, evidence-based trust decisions about nodes, services, and data sources across the continuum without reliance on a central authority.

5.5 Quantum Key Distribution (QKD) for Critical Links: While PQC addresses algorithmic quantum-resistance, QKD provides information-theoretically secure key exchange over optical fibre for point-to-point high-value links (government, critical infrastructure backbone). These two technologies are complementary and together form a comprehensive quantum-safe strategy.

5.6 Secure Multi-Party Computation (SMPC): Homomorphic Encryption and TEEs (Layer 5.1) handle confidential computing for single-party workloads. SMPC enables multiple mu-

tually distrusting parties to jointly compute a function over their combined private inputs without revealing those inputs to each other—a critical capability for cross-border analytics and federated AI in regulated sectors.

Layer 6: Network Infrastructure & Perimeter-Centric Security

6.1 Telco-cloud & O-RAN Security: Hardening distributed 5G/6G cores and RAN components must align with the **Smart Networks and Services Joint Undertaking (SNS JU) SRIA** [5]. This ensures sovereign control over network slice isolation and virtualized RAN components.

6.2 Cross-Sector Physical Resilience & Safety: Adapting strict network controls and IT/OT gateway security for high-availability edge environments (e.g., Smart Grids). This guarantees that cyber-breaches in the digital layer do not cascade into physical safety hazards or kinetic damage in critical infrastructure.

6.3 Edge-Native Autonomous Guardians: Hosting lightweight, localized security control functions (Security-as-a-Service aggregators) directly at the far-edge gateways or within Open-RAN Radio Intelligent Controllers (RIC). These autonomous guardian nodes execute real-time threat mitigation without requiring round-trip communication to a centralized cloud-based SOC, dramatically reducing response latency.

6.4 Non-Terrestrial Network (NTN) & Satellite Security: 6G will natively integrate satellite segments for coverage continuity, backhaul resilience, and emergency operations. These non-terrestrial links introduce distinct threat surfaces—including signal interception and orbital spoofing—requiring dedicated security requirements aligned with 3GPP NTN specifications and the EU Space Programme.

Part B: The Transversal Security Verticals (Cross-Layers)

These four cross-layers represent continuous, automated processes and governance structures that intersect every layer of the vertical stack.

Cross-Layer 1: Automated Security Processes & Cyber-Resilient AI Lifecycle

CL1.1 Continuous DevSecOps (Shift-Left): Embedding AI-driven code analysis and automated vulnerability remediation at the earliest stages of the software lifecycle.

CL1.2 Model & Pipeline Integrity (DataBOMs): Securing AI training pipelines to prevent data poisoning and unauthorized model manipulation.

CL1.3 Runtime AI Defense: Defending AI/ML infrastructure against real-world adversarial attacks (e.g., prompt injection, model evasion/theft) at the edge.

CL1.4 Dynamic Supply Chain Verification (Runtime SBOMs): Moving beyond static software manifests by implementing dynamic Software Bill of Materials (Runtime SBOMs) and “cryptographic proof of execution” to continuously verify the integrity of containerized microservices and AI components as they migrate across the multi-provider edge-cloud continuum.

CL1.5 Automated Adversarial Testing & Continuous Purple Teaming: DevSecOps pipelines focus on code analysis and remediation; the adversarial counterpart provides continuous automated penetration testing, fuzzing (coverage-guided and protocol-level), and purple-team automation that simulates real attacker behaviour against live staging environments, ensuring defenses are validated continuously rather than at periodic intervals.

Cross-Layer 2: Dynamic Risk Management & Operational Cyber Resilience

CL2.1 Automated SOC Operations & Collaborative Threat Intelligence: Continuous monitoring, SOC federation across Member States, and automated incident containment, underpinned by legally protected, cross-border threat intelligence sharing ecosystems designed to overcome institutional trust barriers and incentivize active collaboration [8].

CL2.2 Systemic & Cascading Risk Mitigation: Mapping the evolving threat landscape and using impact-likelihood scoring to dynamically sever links and prevent cross-domain systemic failures.

CL2.3 High-Precision Telemetry and Security Signalling: Implementing “smart telemetry” and cybersecurity-related signalling to feed real-time threat intelligence into analytics platforms without overwhelming network bandwidth—ensuring that visibility does not itself become an attack vector or performance bottleneck.

CL2.4 Internet of Behavior (IoB) and Evidence Sharing: Utilizing cybersecurity and QoS signalling traffic to extract and share behavioral patterns across the infrastructure, establishing an “Internet of Behavior” that enables proactive anomaly correlation and evidence-based risk assessment at continental scale.

CL2.5 AI-Native Analytics Integration (NWDAF & LLMSecOps): Routing cybersecurity signalling traffic and telemetry through centralized analytics hubs powered by AI agents. This creates a live “nervous system” for the network, allowing AI agents to continuously monitor for zero-day exploits, analyze encrypted traffic flows, and trigger automated incident responses aligned with NWDAF (Network Data Analytics Function) standards.

CL2.6 Cyber Threat Intelligence (CTI) Lifecycle Management: While CL2.1 and CL2.4 consume and act on threat data, explicit governance of the full CTI lifecycle—collection, processing, analysis, dissemination, and feedback—is required as a prerequisite for the federated SOC vision. A sovereign EU CTI platform must define this governance layer to ensure data quality, legal compliance, and attribution reliability.

CL2.7 Cyber Deception Technology & Active Defense (Honeynets, Canaries): Current detection is passive—telemetry is observed and anomalies flagged. Deception adds an active intelligence-collection dimension by deploying decoy assets (honeypots, honeynets, credential canaries, fake API endpoints) that generate high-fidelity, low-noise alerts when touched, enabling early detection of advanced intrusions and insider threats.

Cross-Layer 3: Interoperable Security Mechanisms & Open-Source Verifiability

CL3.1 Heterogeneous Interoperability & Simpl Integration: Secure communication and data exchange across disparate systems, edge nodes, and proprietary domains. This includes mandatory alignment with and integration into the Simpl open-source smart middleware [7] to ensure secure, federated access across Common European Data Spaces.

CL3.2 Minimal Interoperability Mechanisms (MIMs) & Open Verifiability: Establishing standardized, baseline security and privacy controls to ensure trust without creating vendor lock-in. This relies heavily on embracing open-source software and open hardware architectures as fundamental prerequisites for transparency, cryptographic verifiability, and the elimination of proprietary black-box vulnerabilities [7].

CL3.3 Neutral Host and Multi-Tenant Isolation: As the ecosystem shifts toward infrastructure sharing and Neutral Host models spanning wireless, optical, and cloud domains, robust

isolation between tenants is critical. Security architectures must enforce Zero-Trust edge operations across different network partitions to prevent cross-tenant data leakage and privilege escalation.

CL3.4 Pan-European NaaS Federation and Cross-Border Edge Roaming: Establishing standardized interfaces and harmonized Network APIs (e.g., leveraging CAMARA) to interconnect the fragmented edge clouds of different European telecom operators into a unified Network-as-a-Service (NaaS) platform—enabling a “Schengen for Cloud/AI” where compute workloads can roam seamlessly across borders.

CL3.5 Semantic Security Interoperability (Common Ontologies & Data Models): Federated SOCs and shared threat intelligence only function if participating organisations share a common semantic understanding of security events, asset taxonomies, and risk classifications. Standardized ontologies (e.g., aligned with STIX/TAXII and MITRE ATT&CK) are a prerequisite for meaningful cross-border correlation.

CL3.6 Cyber Range & Digital Twin Federation for Security Validation: Before deploying cross-border federated security configurations into production, they must be validated in high-fidelity simulation environments. ENISA already coordinates national cyber range inventories; this cross-layer would federate them into a shared EU-wide infrastructure aligned with the Cyber Solidarity Act’s testing provisions.

Cross-Layer 4: Compliance-by-Design, Agile Governance & Automated Assurance

CL4.1 Policy-as-Code & Compliance-by-Design: Technical implementation of legal requirements (OSCAL) into machine-readable rules that pre-validate workloads.

CL4.2 Continuous Automated Auditing & Certification Lifecycle: Replacing “point-in-time” PDF certificates with real-time, telemetry-driven certification tracking for both cloud services (EUCS) [4] and physical ICT edge products (EUCC) [11]. This ensures that edge nodes autonomously maintain compliance by integrating mandatory vulnerability handling and cryptographic patch verification directly into their operational lifecycle, strictly aligning with EUCC maintenance and assurance rules [11].

CL4.3 Post-Quantum Readiness & Zero-Trust Architectures (ZTA): Moving beyond traditional perimeter-based defenses by embedding continuous, AI-driven Zero-Trust verification and micro-segmentation natively into the cloud-edge infrastructure, combined with a systematic migration roadmap to post-quantum cryptographic primitives across all layers.

CL4.4 Continuous Automated Certification: Replacing “point-in-time” PDF certificates with real-time, telemetry-driven certification tracking for both cloud services (EUCS) and physical ICT edge products (EUCC), ensuring that edge nodes autonomously maintain compliance by integrating mandatory vulnerability handling and cryptographic patch verification directly into their operational lifecycle [11].

CL4.5 Multicloud Key Management as a Service (KMaaS): When AI workloads, data flows, and context flows migrate dynamically across a multi-provider computing continuum, managing cryptographic keys and access policies becomes highly complex. KMaaS provides a provider-agnostic, sovereign key lifecycle management capability that prevents key sprawl and ensures revocation propagates consistently across heterogeneous environments.

CL4.6 Privacy-Preserving Federated Computation: When multiple providers work on the same infrastructure, AI training and inference may require access to datasets stored across different jurisdictions and competitive boundaries. Privacy-preserving techniques (federated learning, differential privacy, SMPC) enable collaborative computation while guaranteeing that raw data

never leaves its sovereign jurisdiction.

CL4.7 Cyber Risk Quantification & Board-Level Reporting (CRQ): Compliance frameworks define what controls to implement. A dedicated cross-layer translates technical security posture into financial and operational risk language for decision-makers—potentially in real-time—enabling boards to make informed, evidence-based investment and risk-acceptance decisions aligned with DORA requirements.

CL4.8 AI Trustworthiness & Algorithmic Accountability: As AI systems themselves become actors within the security architecture—making autonomous containment decisions, classifying threats, orchestrating responses—the trustworthiness of those AI systems becomes a security-critical concern. This layer defines explainability, auditability, and override mechanisms for AI-driven security functions, ensuring alignment with the EU AI Act for high-risk systems.

CL4.9 Regulatory Digital Twins: Virtualized replicas of critical infrastructure specifically to stress-test regulatory compliance against evolving laws before applying changes to the live physical continuum.

7 Priorities

7.1 Executive Priorities: The Critical Path (Phase 1)

Applying the 4-Filter Framework, the following areas are identified as immediate strategic priorities for the 2025-2030 funding cycle:

- **Priority 1:** Hardware Roots of Trust & EUCC 'High' Certification (Filters 1, 2, 3).
- **Priority 2:** Post-Quantum Cryptographic Agility for Edge/IoT (Filter 3).
- **Priority 3:** Lightweight Cryptography & Physical-Layer Security (PhySec) for the Extreme Edge.
- **Priority 4:** Security Interoperability via Simpl Middleware (Filter 4).
- **Priority 5:** Supply Chain Verification and Secured Infrastructure.
- **Priority 6:** Cognitive SecOps (LLMSecOps) & Automated Threat Intelligence.
- **Priority 7:** Pan-European NaaS Federation ("Schengen for Cloud-Edge/AI").
- **Priority 8:** Confidential Computing & Privacy-Preserving AI Pipelines.
- **Priority 9:** Resilient-by-Design Infrastructure & Autonomous "Island Mode."
- **Priority 10:** Proactive Cyber Threat Management & Defense against Advanced Persistent Threats (APTs).

8 Capability Profiles & Progression Matrices

Priority 1: Hardware Roots of Trust & EUCC 'High' Certification

1. **Baseline & Target Objective:** *Baseline:* High foreign dependency on proprietary Trusted Platform Modules (TPMs) and secure elements. Far-edge devices lack standardized hardware anchors. *Target:* Ubiquitous deployment of EU-designed, open-source (RISC-V) secure enclaves certified at **EUCC 'High' assurance**, providing a mathematically verifiable foundation for the entire continuum.
2. **Primary Drivers:** *Threat:* State-level physical tampering and side-channel attacks on decentralized edge nodes. *Regulatory:* Mandatory hardware security requirements under the **Cyber Resilience Act (CRA)** and certified anchors for **NIS2** critical infrastructure.
3. **High Impact Activities:** Development of sovereign European Secure Elements based on RISC-V; automated hardware-level remote attestation protocols for large-scale edge fleets; EU-level standard toolset developed and maintained by the European Commission; SME support for certification and compliance testing; security-based sensor research for future deployment.
4. **Catalysts & Enablers:** **EU Chips Act** pilot lines for secure silicon; **ECCC Strategic Agenda** funding for open-hardware security; alignment with CEN/CENELEC hardware security standards.

Metric	Phase 1: Foundation & Compliance (0-5 Yrs)	Phase 2: Federation & Standardization (5-10 Yrs)	Phase 3: Autonomy & Leadership (10-15 Yrs)
ODP	Fragmented Piloting	Standardized Integration	Ubiquitous Operation
SSL	High Foreign Dependency	Emerging EU Ecosystem	Full Sovereign Autonomy
Operational State	Cloud-based TPMs common; Edge nodes lack hardware trust; RISC-V pilots begin; CRA compliance starts.	Standardized EU-made secure elements in critical infra; Automated attestation at scale.	Sovereign hardware trust embedded by default in all EU ICT products from IoT to HPC.
Critical Enabler	Cyber Resilience Act (CRA) enforcement	EUCC 'High' certification of EU-based secure elements	Maturity of EU Chips Act sovereign fabrication lines

5. **Disruptors & Systemic Risks:** Supply chain bottlenecks for EU-based fabrication; breakthroughs in physical side-channel analysis that bypass current cryptographic enclaves.
6. **Failsafe Mechanisms & Resilience:** “Defense-in-depth” where software-based isolation (TEEs) acts as a secondary layer if a hardware enclave is compromised; autonomous node isolation and distributed attestation if hardware integrity is lost.
7. **Transversal Meta-Layer:** *Sovereignty:* Eliminates “black-box” hardware dependencies on non-EU vendors. *Sustainability:* Energy-efficient secure silicon designs reducing the carbon footprint of cryptographic operations at the far-edge.

Progression Matrix: Hardware Roots of Trust & EUCC 'High'

Priority 2: Post-Quantum Cryptographic (PQC) Agility for Edge/IoT

1. **Baseline & Target Objective:** *Baseline:* Current IoT and edge infrastructure predominantly relies on lightweight cryptography primitives which can be vulnerable to future quantum computers. Limited awareness and no systematic migration path for resource-constrained devices. *Target:* A “Quantum-Native” continuum where all critical edge/IoT infrastructure has migrated to hybrid or full Post-Quantum Cryptography, with automated crypto-agility enabling seamless algorithm updates without hardware replacement.
2. **Primary Drivers:** *Threat:* The **Q-Day** threat and “Harvest Now, Decrypt Later” strategies targeting sensitive long-lived data. *Regulatory:* Emerging ENISA guidelines and future EU mandates requiring PQC readiness for critical infrastructure under NIS2.
3. **High Impact Activities:** Optimization of NIST-standardized PQC algorithms for 8/16-bit microcontrollers; development of crypto-agility middleware for OTA algorithm updates; automated cryptographic inventory and migration tools for legacy IoT deployments; 3GPP specifications alignment for quantum key distribution (QKD) in cellular systems.
4. **Catalysts & Enablers:** Post-Quantum Cryptography standardization initiatives; **3GPP** specifications for quantum key distribution (QKD); **NIST PQC standardization finalization**; **ETSI** quantum-safe cryptography standards; **ENISA PQC guidelines**; **ECCC Strategic Agenda** funding; EU Chips Act for PQC-native silicon development.

Metric	Phase 1: Foundation & Compliance (0-5 Yrs)	Phase 2: Federation & Standardization (5-10 Yrs)	Phase 3: Autonomy & Leadership (10-15 Yrs)
ODP	Applied R&D	Fragmented Piloting	Ubiquitous Operation
SSL	High Foreign Dependency	Emerging EU Ecosystem	Full Sovereign Autonomy
Operational State	NIST algorithm selection; software-based PQC pilots in cloud environments; research on lightweight implementations.	Hybrid (classical + PQC) schemes in Industrial IoT and critical edge infrastructure; EU-developed crypto-agility middleware.	Fully quantum-resilient continuum with PQC embedded in all devices; automated crypto-agility across Cloud-Edge-IoT.
Critical Enabler	ENISA PQC Guidelines and NIST standardization completion	NIST/ETSI algorithm finalization and EU-wide deployment standards	PQC-native RISC-V silicon and ubiquitous crypto-agility frameworks

5. **Disruptors & Systemic Risks:** Early arrival of CRQC before migration is complete; performance overhead of PQC algorithms on battery-powered IoT; fragmented standardization; insufficient cryptographic inventory of legacy OT/IoT systems.
6. **Failsafe Mechanisms & Resilience:** Deployment of hybrid cryptographic schemes (classical + PQC) during the transition; crypto-agility frameworks allowing rapid algorithm switching if vulnerabilities are discovered; diversification of PQC algorithm families to avoid single-point cryptanalytic failures.
7. **Transversal Meta-Layer:** *Sovereignty:* Protection of Europe’s long-term industrial and governmental secrets from future quantum decryption. *Sustainability:* Energy-efficient, lightweight PQC implementations reducing the carbon footprint of security (Green IT).

Progression Matrix: PQC Agility for Edge/IoT

Priority 3: Lightweight Cryptography & Physical-Layer Security (PhySec) for the Extreme Edge

1. **Baseline & Target Objective:** *Baseline:* Massive IoT and nano-devices cannot support computationally heavy classical asymmetric encryption. Current security approaches are unsuitable for environments requiring 100 devices per m³ with 40-year battery lifespans or zero-energy backscatter radios. *Target:* Move to ultra-lightweight Post-Quantum Cryptography and keyless Physical-Layer Security (PhySec) embedded at the extreme edge, enabling security primitives that operate within severe energy and compute constraints.
2. **Primary Drivers:** *Threat:* The proliferation of massive Machine Type Communication (mMTC) creates an enormous unprotected attack surface; classical cryptography is both computationally infeasible and imminently vulnerable to quantum algorithms at the extreme edge. *Regulatory:* CRA requirements extend to the smallest connected devices; NIS2 mandates security across the full infrastructure including resource-constrained nodes.
3. **High Impact Activities:** Group-based authentication for large-scale constrained IoT fleets; RF fingerprinting and Channel State Information (CSI)-based device authentication; integration of lightweight PQC into European open-hardware (RISC-V) for smart body-area net-

Metric	Phase 1: Foundation & Compliance (0-5 Yrs)	Phase 2: Federation & Standardization (5-10 Yrs)	Phase 3: Autonomy & Leadership (10-15 Yrs)
ODP	Isolated Lightweight Piloting	Standardized D2D Security Integration	Ubiquitous Zero-Energy Trust
SSL	High Dependency on Legacy/Heavy Algorithms	Emerging EU Lightweight & PhySec Ecosystem	Full Sovereign Extreme-Edge Cryptography
Operational State	Group-based authentication and early lightweight PQC algorithms tested on resource-constrained IoT; basic RF fingerprinting used for anomaly detection.	Lightweight PQC and PhySec natively embedded in European open-hardware (RISC-V) and smart body-area networks (WBAN).	Battery-less (zero-energy) nodes autonomously execute quantum-resistant authentication and trusted decentralized swarm communications.
Critical Enabler	NIST Lightweight PQC Standards & Context-Aware IoT Platforms	Hardware-Software Co-Design for Neuromorphic/PIM AI	“Zero-Energy” Air Interfaces & Ubiquitous ISAC (Integrated Sensing and Communication)

works (WBAN); ETSI SmartBAN standardization alignment; research into “zero-energy” air interfaces with integrated Integrated Sensing and Communication (ISAC) security.

4. **Catalysts & Enablers:** NIST lightweight post-quantum cryptography standardization; ETSI SmartBAN standardization; 6G research into RF fingerprinting and channel entropy for native security; Hardware-Software Co-Design for neuromorphic and Processing-in-Memory (PIM) architectures.
5. **Disruptors & Systemic Risks:** Performance gap between available energy budgets and minimum cryptographic processing requirements; fragmented device ecosystems preventing uniform PhySec adoption; lack of EU-manufactured ultra-constrained secure silicon.
6. **Failsafe Mechanisms & Resilience:** Dynamic fallback to localized, non-routed trust boundaries leveraging purely physical-layer signatures (RF fingerprinting or channel entropy) to authenticate devices if upper-layer cryptographic processing exceeds the available energy budget or is compromised.
7. **Transversal Meta-Layer:** *Sovereignty:* Sovereign control over the security of the most proliferated and currently most unprotected layer of the computing continuum. *Sustainability:* Zero-energy security primitives are inherently aligned with Green IT objectives and circular economy principles.

Progression Matrix: Lightweight Cryptography & PhySec for the Extreme Edge

Priority 4: Security Interoperability via Simpl Middleware

1. **Baseline & Target Objective:** *Baseline:* Current European cloud and edge infrastructure is highly fragmented across proprietary vendor silos with incompatible security policies, authentication mechanisms, and data exchange protocols. This fragmentation prevents seamless data flow across Common European Data Spaces and creates vendor lock-in. *Target:* A unified European “Trust Fabric” enabling secure, federated interoperability across heterogeneous

cloud and edge providers through mandatory integration with the **Simpl** open-source smart middleware stack, with standardized security Minimal Interoperability Mechanisms (MIMs).

2. **Primary Drivers:** *Regulatory: Data Act* mandates requiring seamless data portability and interoperability; requirements for Common European Data Spaces to operate across multiple providers; NIS2 requirements for coordinated incident response. *Market:* Breaking vendor lock-in to enable European SMEs to compete; enabling cross-border data flows while maintaining sovereignty.
3. **High Impact Activities:** Implementation and standardization of security-specific MIMs within the Simpl stack; automated policy translation engines converting provider-specific security policies into common machine-readable format (XACML, OSCAL); federated identity bridges using verifiable credentials; cross-domain security orchestration for unified threat response.
4. **Catalysts & Enablers:** **Digital Europe Programme** funding; **Simpl** open-source stack community and governance; **Gaia-X** federation services and trust frameworks; ECCC support for security MIM standardization.
5. **Disruptors & Systemic Risks:** Vendor resistance to open standards; weakest-link security failures in federated systems; complexity of policy translation creating semantic gaps; fragmented national implementations creating “data islands.”
6. **Failsafe Mechanisms & Resilience:** Capability to autonomously “unfederate” or isolate nodes that fail to meet the shared security baseline; zero-trust architectures where federation does not imply implicit trust; continuous compliance monitoring with automated ejection of non-compliant providers.
7. **Transversal Meta-Layer:** *Sovereignty:* Breaking dependence on non-EU hyperscaler lock-in. *Trust:* Fostering pan-European cross-border trust through transparent, verifiable security mechanisms. *Economic Viability:* Enabling SMEs to participate in Data Spaces without massive infrastructure investments.

Progression Matrix: Security Interoperability via Simpl Middleware

Priority 5: Supply Chain Verification and Secured Infrastructure

1. **Baseline & Target Objective:** *Baseline:* Supply chain tracking is static, opaque, and fragmented. The disaggregation of telco clouds (e.g., Open RAN) introduces extreme complexity by mixing software and hardware from multiple vendors. SMEs lack resources to meet CRA and NIS2 supply chain compliance requirements. *Target:* Move to continuously verified ecosystems utilizing Runtime Software Bill of Materials (SBOMs), Digital Product Passports (DPP), and automated certification across multi-vendor deployments, with full supply chain transparency and sovereign control.
2. **Primary Drivers:** *Threat:* Sophisticated supply chain attacks (e.g., SolarWinds-style compromises) targeting the disaggregated Open RAN and cloud-edge ecosystem; hidden backdoors in non-EU hardware and firmware. *Regulatory:* Strict legal mandates from the **Cyber**

Metric	Phase 1: Foundation & Compliance (0-5 Yrs)	Phase 2: Federation & Standardization (5-10 Yrs)	Phase 3: Autonomy & Leadership (10-15 Yrs)
ODP	Fragmented Piloting	Standardized Integration	Ubiquitous Operation
SSL	Emerging EU Ecosystem	Full Sovereign Autonomy	Full Sovereign Autonomy
Operational State	Initial Simpl security pilots in select Data Spaces; manual policy coordination; proof-of-concept MIM implementations.	Automated policy translation via standardized MIMs; Simpl-based security in majority of Common Data Spaces; cross-border federated SOC operations.	Fully automated, provider-agnostic European trust fabric; real-time security orchestration across the continuum; zero-trust federation by default.
Critical Enabler	Simpl-Open official launch and initial MIM definitions	Security MIMs adoption as CEN/CENELEC standards	EU Cloud Rulebook maturity and Data Act full enforcement

Resilience Act (CRA) and **NIS2 Directive** demanding robust supply chain risk management; Digital Product Passport (DPP) initiatives for traceability.

3. **High Impact Activities:** Review and mapping of hardware, OS, and software supply chain vulnerabilities; SME engagement prioritized—ensuring one-size-fits-all approaches are avoided and SMEs can enable new technology integration (e.g., NXP ecosystems); Dynamic Runtime SBOMs tracking active component states; Digital Product Passports for end-to-end traceability; Zero-Knowledge Proofs (ZKPs) for IP-protected component verification.
4. **Catalysts & Enablers:** Digital Product Passport (DPP) initiatives; European financial support for SME security upskilling; ECCC strategic funding for supply chain security tooling; Pan-European threat intelligence sharing for supply chain indicators of compromise.
5. **Disruptors & Systemic Risks:** Extreme complexity of multi-vendor Open RAN supply chains; geopolitical dependencies on non-EU component manufacturers; SMEs being priced out of compliance; lack of standardized machine-readable supply chain attestation formats.
6. **Failsafe Mechanisms & Resilience:** Graceful isolation of unverified or high-risk components into quarantined network slices until cryptographic proofs of execution can validate their integrity; automated blocking of unverified components in production deployments.
7. **Transversal Meta-Layer:** *Sovereignty:* Reducing systemic dependency on non-EU hardware and software supply chains. *Trust:* Providing verifiable, transparent provenance for all components in the European computing continuum. *Economic:* Protecting EU industrial IP through privacy-preserving verification mechanisms.

Progression Matrix: Supply Chain Verification and Secured Infrastructure

Priority 6: Cognitive SecOps (LLMSecOps) & Automated Threat Intelligence

1. **Baseline & Target Objective:** *Baseline:* Security Operations Centers (SOCs) are predominantly manual, reactive, and siloed. Analyst capacity is far outpaced by the expanding

Metric	Phase 1: Foundation & Compliance (0-5 Yrs)	Phase 2: Federation & Standardization (5-10 Yrs)	Phase 3: Autonomy & Leadership (10-15 Yrs)
ODP	Full Supply Chain Disclosure	Partial Restriction Process	Full Restriction Process
SSL	High Reliance on Fragmented, Non-EU Supply Chain Auditing	Emerging EU Supply Chain Tracking Frameworks (DPP)	Full Sovereign Control and Supply Chain Transparency
Operational State	Review of hardware, OS, and software supply chain vulnerabilities; initial Runtime SBOM pilots; CRA compliance mapping begins.	Ensuring certification of critical supply chain hardware, OS, and software; DPP frameworks operational; SME compliance pathways established.	Fully certified supply chain actively blocking unverified components; ZKPs enable IP-protected verification at continental scale.
Critical Enabler	CRA supply chain requirements enforcement; SME engagement programme	Dynamic Runtime SBOMs and Digital Product Passports	Zero-Knowledge Proofs (ZKPs) & Pan-European Threat Intel sharing

attack surface of 5G/6G and the millions of IoT devices. EU SOC's are heavily reliant on non-EU AI foundation models. *Target:* Autonomous, AI-driven “LLMSecOps” agents distributed across the 6G edge, capable of proactive threat hunting, real-time incident containment, and continuous learning without requiring human intervention for routine responses.

2. **Primary Drivers:** *Threat:* The dual-use nature of AI generating complex offensive threats (APTs, data poisoning, AI-driven exploit generation); the expanding attack surface of 6G and millions of IoT devices overwhelming human-scale SOC operations. *Regulatory:* EU AI Act compliance for high-risk autonomous security systems; Zero-Touch Network & Service Management (ZSM) goals; EuroHPC AI Factories as the compute substrate for sovereign AI models.
3. **High Impact Activities:** LLM-assisted human SOC analysts integrated with NWDAF telemetry; edge-native “Guardian” nodes executing Zero-Trust reconfigurations autonomously; European sovereign Small Language Models (SLMs) for security classification; swarm-based agentic AI for decentralized threat intelligence; standardized AI Interconnect APIs.
4. **Catalysts & Enablers:** EU AI Act compliance frameworks for high-risk security AI; EuroHPC AI Factories providing compute for sovereign foundation model training; ECCC Strategic Agenda for LLMSecOps standardization; NWDAF integration with 3GPP standards.
5. **Disruptors & Systemic Risks:** AI hallucination causing false containment actions disrupting critical services; adversarial attacks on the AI models themselves (prompt injection, model poisoning); regulatory uncertainty around autonomous AI decision-making in security contexts; continued dominance of non-EU foundation models.
6. **Failsafe Mechanisms & Resilience:** “Shadow-mode” rollback and human-in-the-loop (RLHF) intervention if autonomous edge agents hallucinate or exhibit unstable control loops; mandatory override mechanisms for all autonomous security decisions affecting critical infrastructure.
7. **Transversal Meta-Layer:** *Sovereignty:* Development of European sovereign edge LLMs and SLMs replacing dependency on US foundation models. *Trust:* Ensuring that AI-driven

Metric	Phase 1: Foundation & Compliance (0-5 Yrs)	Phase 2: Federation & Standardization (5-10 Yrs)	Phase 3: Autonomy & Leadership (10-15 Yrs)
ODP	Siloed AI Copilots	Federated Edge Agents	Ubiquitous AI Immune System
SSL	High Reliance on US Foundation Models	Sovereign European Edge LLMs, SLMs, LCMs	Global Leadership in Trustworthy AI
Operational State	LLMs assist human SOC analysts; NWDAF telemetry routed to centralized AI for anomaly detection.	Edge-native “Guardian” nodes autonomously execute Zero-Trust reconfigurations and throttle threats.	Swarm-based agentic AI defends the Edge-Cloud Continuum via decentralized, proactive threat intelligence.
Critical Enabler	EU AI Act enforcement & EuroHPC AI Factories	Standardized AI Interconnect APIs	Self-Evolving AI Swarm Protocols

security decisions are explainable, auditable, and aligned with the EU AI Act for high-risk systems. *Sustainability*: Energy-efficient edge AI inference reducing the carbon footprint of autonomous security operations.

Progression Matrix: Cognitive SecOps (LLMSecOps) & Automated Threat Intelligence

Priority 7: Pan-European NaaS Federation (“Schengen for Cloud-Edge/AI”)

1. **Baseline & Target Objective:** *Baseline*: European telco deployments are fragmented and nation-bound. The lack of a true Digital Single Market prevents European operators from scaling to compete with US and Chinese hyperscalers. *Target*: A unified, interoperable European Network-as-a-Service (NaaS) platform enabling seamless “roaming for compute” across borders, legally operating as a single jurisdiction for data workloads.
2. **Primary Drivers:** *Market*: The “Sovereignty Cliff”—lack of a true Digital Single Market prevents European operators from achieving scale parity with hyperscalers. *Regulatory*: Data Act interoperability requirements; IPCEI-CIS mandates for cross-border infrastructure investment; CAMARA API standardization enabling federated network services.
3. **High Impact Activities:** Deployment of harmonized CAMARA APIs; automated compliance negotiation in 5G corridors; “Data Visa” mechanisms enabling dynamic cross-border AI model migration; inter-operator East-West security interfaces; development of the “Schengen for Cloud” legal framework.
4. **Catalysts & Enablers:** IPCEI-CIS initial deployments; **European Alliance for Industrial Data, Edge and Cloud**; **CAMARA API alliance**; **Gaia-X** federation services; **ECCC** for cross-border security standards.
5. **Disruptors & Systemic Risks:** Conflicting national data sovereignty laws preventing true federation; regulatory arbitrage creating security-weakest-link vulnerabilities at cross-border interfaces; insufficient SLA enforcement mechanisms across operator boundaries.
6. **Failsafe Mechanisms & Resilience:** Graceful degradation protocols redirect workloads to national infrastructure if cross-border SLAs or compliance APIs fail; bilateral fallback agreements between operator pairs ensuring minimum viable connectivity.

Metric	Phase 1: Foundation & Compliance (0-5 Yrs)	Phase 2: Federation & Standardization (5-10 Yrs)	Phase 3: Autonomy & Leadership (10-15 Yrs)
ODP	Bilateral Interconnection	Federated Edge Roaming	True Digital Single Market
SSL	Hyperscaler Platform Dominance	Pan-European Consortium Competes	European Scale Parity Achieved
Operational State	Harmonized CAMARA APIs deployed; Automated Compliance Negotiators tested in 5G corridors.	“Data Visas” enable dynamic cross-border migration of AI models without legal friction.	Single unified computing continuum legally operates as one jurisdiction for data workloads.
Critical Enabler	IPCEI-CIS Initial Deployments	Inter-Operator East-West Interfaces	“Schengen for Cloud” Legal Framework

7. **Transversal Meta-Layer: *Sovereignty*:** Enabling European-scale infrastructure that competes with US and Chinese hyperscalers without ceding control to non-EU providers. *Trust*: “Data Visas” providing legal clarity on data jurisdiction during cross-border migration. *Economic*: Creating a genuine Digital Single Market that enables European SMEs and operators to compete at continental scale.

Progression Matrix: Pan-European NaaS Federation

Priority 8: Confidential Computing & Privacy-Preserving AI Pipelines

1. **Baseline & Target Objective:** *Baseline*: Basic encryption at-rest and in-transit is standard, but data-in-use remains exposed during processing. Multi-tenant edge nodes and federated AI training create significant insider threat and IP leakage risks. *Target*: Ubiquitous data-in-use protection via Trusted Execution Environments (TEEs) and Homomorphic Encryption across the full Cloud-to-IoT path, enabling collaborative AI without raw data ever leaving sovereign jurisdictions.
2. **Primary Drivers:** *Threat*: Insider threats in multi-tenant edge nodes; IP exfiltration during federated AI training across competitive organizational boundaries. *Regulatory*: **GDPR** stringency for data-in-use; **EUCS “High+”** certification requirements mandating confidential computing for sensitive cloud workloads; **AI Act** requirements for data governance in high-risk AI systems.
3. **High Impact Activities:** TEE deployment on critical cloud nodes for EUCS compliance; sovereign hardware pilots using EU-developed RISC-V TEEs; multi-party privacy-preserving AI pipelines across multi-provider nodes; operationalization of Fully Homomorphic Encryption (FHE) for production analytics workloads.
4. **Catalysts & Enablers:** **EUCS** Certification Rollout; **European Data Spaces** as the deployment vehicle; **Chips JU** for RISC-V hardware development; **ECCC** strategic funding for FHE maturation.
5. **Disruptors & Systemic Risks:** TEE side-channel vulnerabilities rendering current enclave implementations obsolete; FHE performance overhead remaining prohibitive for real-time edge applications; continued dependency on non-EU TEE hardware (Intel SGX, ARM TrustZone).

Metric	Phase 1: Foundation & Compliance (0-5 Yrs)	Phase 2: Federation & Standardization (5-10 Yrs)	Phase 3: Autonomy & Leadership (10-15 Yrs)
ODP	Isolated Enclaves	Distributed Confidential Cloud	Ubiquitous Confidential Continuum
SSL	High Reliance on Foreign Hardware TPMs	European RISC-V TEEs Emerge	Total Data-in-Use Sovereignty
Operational State	TEEs deployed on critical cloud nodes for EUCS compliance; initial sovereign hardware pilots.	Multi-party privacy-preserving AI pipelines train across multi-provider nodes seamlessly.	Homomorphic encryption operationalized; end-to-end encrypted continuum from IoT to HPC.
Critical Enabler	EUCS Certification Rollout	RISC-V Hardware Scaling	Fully Homomorphic Encryption (FHE) Maturation

6. **Failsafe Mechanisms & Resilience:** Limit processing to local, on-device federated learning nodes (never sharing raw data) if edge TEE integrity cannot be cryptographically verified; multi-TEE implementation diversity to avoid single-vendor hardware vulnerabilities.
7. **Transversal Meta-Layer:** *Sovereignty:* Moving from reliance on foreign hardware TEEs to sovereign RISC-V TEEs. *Trust:* Enabling European citizens and businesses to share data for AI without sacrificing privacy. *Ethics:* Providing the technical foundation for Privacy-by-Design as required by GDPR.

Progression Matrix: Confidential Computing & Privacy-Preserving AI Pipelines

Priority 9: Resilient-by-Design Infrastructure & Autonomous “Island Mode”

1. **Baseline & Target Objective:** *Baseline:* Current cloud-dependent architectures have single points of failure. Edge nodes cannot function autonomously if WAN connectivity or centralized cloud services are disrupted. NIS2 mandates business continuity but infrastructure lacks native resilience capabilities. *Target:* Edge-native architectures capable of graceful degradation and autonomous local operation (“Island Mode”) during severe network, cloud, or power disruptions—ensuring continuity of critical services without any central cloud intervention.
2. **Primary Drivers:** *Threat:* Escalating global stressors—cyberattacks, extreme weather, geopolitical tensions—causing cascading failures across interconnected cloud-edge infrastructure. *Regulatory:* **NIS2** mandates for business continuity in critical infrastructures (healthcare, energy, transport); **DORA** (Digital Operational Resilience Act) requirements for financial sector; SNS JU requirements for 6G resilience and AI-RAN.
3. **High Impact Activities:** Edge node soft-state protocols enabling operation with cached credentials during cloud disconnections; AI-driven anomaly detection triggering graceful degradation automatically; dynamic spectrum sharing for emergency roaming; zero-energy backup layers maintaining minimal vital telemetry; bio-inspired elasticity for autonomous E2E recovery.

Metric	Phase 1: Foundation & Compliance (0-5 Yrs)	Phase 2: Federation & Standardization (5-10 Yrs)	Phase 3: Autonomy & Leadership (10-15 Yrs)
ODP	Localized Fallback Piloting	Standardized Island Mode	Fully Autonomous Cyber-Physical Resilience
SSL	Vulnerable to Cascading WAN Failures	Contained Local Survivability	Systemic European Resilience
Operational State	Edge nodes use cached credentials and soft-state protocols to survive temporary cloud disconnections.	AI-driven anomaly detection triggers graceful degradation; dynamic spectrum sharing used for emergency roaming.	Bio-inspired elasticity; swarms seamlessly coordinate E2E cyber-physical recovery without any central cloud intervention.
Critical Enabler	NIS2 & DORA Enforcement	3GPP/ETSI Standardized Fallback APIs	“Cognitive Twin” E2E Orchestration

4. **Catalysts & Enablers:** NIS2 & DORA Enforcement as market driver; 3GPP/ETSI standardized fallback APIs; SNS JU 6G resilience research; O-RAN distributed open interfaces enabling autonomous RAN operation.
5. **Disruptors & Systemic Risks:** Security vulnerabilities introduced by offline-mode cached credentials being exploited; lack of standardized Island Mode protocols preventing interoperability during coordinated multi-site disruptions; power infrastructure failures disabling edge nodes before Island Mode activates.
6. **Failsafe Mechanisms & Resilience:** Zero-energy backup layers or localized Low-Power WAN (LPWAN) micro-slices maintaining a “minimal viable network” for life-saving telemetry if standard edge power and computing completely fail; local cryptographic credential caches with time-bounded validity to prevent indefinite offline operation.
7. **Transversal Meta-Layer:** *Sovereignty:* Ensuring European critical infrastructure can operate independently of any centralized or foreign-hosted cloud dependency. *Trust:* Guaranteeing continuity of essential services for European citizens during crises. *Sustainability:* Ultra-low-power Island Mode operation aligning resilience with Green IT objectives.

Progression Matrix: Resilient-by-Design Infrastructure & Autonomous “Island Mode”

Priority 10: Proactive Cyber Threat Management & Defense against Advanced Persistent Threats (APTs)

1. **Baseline & Target Objective:** *Baseline:* Intrusion Detection Systems are reactive with long “dwell times,” allowing APTs to persist undetected for months. Threat intelligence is siloed, national, and non-standardized. Incident response relies heavily on manual analyst workflows. *Target:* Proactive, AI-driven cyber threat hunting, multi-layered alert correlation, and dynamic Moving Target Defense (MTD) to disrupt the prolonged lifecycles of state-sponsored APTs before they achieve their objectives.

2. **Primary Drivers:** *Threat:* The increasing sophistication of nation-state actors blending cyber espionage with organized crime; exploitation of complex supply chains (e.g., SolarWinds-style attacks); use of stealthy techniques like memory-only execution and legitimate administrative tools (“living off the land”) for lateral movement. *Regulatory:* NIS2 incident reporting timelines requiring faster detection and response; ECCC mandate for Pan-European Cyber Threat Intelligence (CTI) sharing.
3. **High Impact Activities:** SIEM/ML correlation of seemingly isolated events to reduce APT dwell times; machine-assisted threat hunting for Indicators of Compromise (IoCs) aligned with **MITRE ATT&CK**; LLMSecOps agents autonomously ingesting CTI and tracking anomalous lateral movement; AI-driven Moving Target Defense (MTD) continuously shuffling network resources and IP addresses to disrupt reconnaissance.
4. **Catalysts & Enablers:** **MITRE ATT&CK** framework integration into automated incident management; Machine Learning and **User and Entity Behavior Analytics (UEBA)** for continuous threat hunting; Pan-European CTI sharing platforms (CONCORDIA successor); **ECCC** coordination of national cyber intelligence capacities.
5. **Disruptors & Systemic Risks:** APT actors adapting faster than defensive AI can learn, using adversarial techniques to evade ML-based detection; legal barriers to real-time CTI sharing across Member State jurisdictions; AI-generated malware bypassing behavioral baselines trained on historical attack patterns.
6. **Failsafe Mechanisms & Resilience:** Autonomous, real-time micro-segmentation (Zero-Trust) to instantly quarantine compromised supply chain components or sever Command and Control (C&C) pathways as soon as behavioral anomalies are detected, even if the specific zero-day exploit is unknown.
7. **Transversal Meta-Layer:** *Sovereignty:* Building European sovereign threat intelligence capacity reducing dependency on non-EU threat feeds and analysis platforms. *Trust:* Protecting European public institutions, critical infrastructure, and democratic processes from state-sponsored cyber operations. *Resilience:* Ensuring that even the most sophisticated adversaries face a continually shifting, proactive defense rather than a static target.

Progression Matrix: Proactive Cyber Threat Management & Defense against APTs

Metric	Phase 1: Foundation & Compliance (0-5 Yrs)	Phase 2: Federation & Standardization (5-10 Yrs)	Phase 3: Autonomy & Leadership (10-15 Yrs)
ODP	Multi-Layered Alert Correlation & Hunting	Automated Triage & LLM-SecOps Containment	Ubiquitous Moving Target Defense (MTD)
SSL	High Dependency on Global Threat Feeds	Sovereign Pan-European CTI & Behavioral Analytics	Full Autonomous Counter-Adversarial Resilience
Operational State	SIEM systems and ML correlate isolated events to reduce APT dwell times; analysts conduct machine-assisted threat hunting for IoCs.	LLMsecOps agents autonomously ingest CTI, track anomalous lateral movement, and orchestrate rapid incident triage aligned with MITRE ATT&CK.	AI-driven MTD continuously shuffles network resources, IP addresses, and software diversity, actively disrupting APT reconnaissance and paralyzing attacks.
Critical Enabler	Federated SIEMs & Standardized CTI Sharing Protocols	LLMsecOps & UEBA (User and Entity Behavior Analytics)	Adaptive Edge-Cloud AI Swarms & Zero-Touch Automation

9 Executive Summary & Strategic Recommendations

Executive Summary

This cybersecurity and cyber-resilience strategy defines a comprehensive 15-year trajectory to secure the European Cloud-Edge-IoT Continuum while establishing digital sovereignty. Applying the **4-Filter Prioritization Framework** (Regulatory Urgency, Foundational Dependency, Sovereignty Cliff, Market Feasibility), ten strategic priorities are identified across three phases following an R&I maturity arc: near-term applied engineering and compliance activation (Phase 1), medium-term applied and industrial R&D with federation and standardization (Phase 2), and long-term frontier research combined with sovereign operational leadership (Phase 3).

The ten priorities are: **P1** Hardware Roots of Trust & EUCC ‘High’; **P2** PQC Agility; **P3** Lightweight Cryptography & PhySec; **P4** Simpl Security Interoperability; **P5** Supply Chain Verification; **P6** Cognitive SecOps / LLMsecOps; **P7** NaaS Federation; **P8** Confidential Computing; **P9** Resilient Infrastructure / Island Mode; **P10** Proactive APT Defense. P1, P2, and P4 constitute the critical-path foundational anchors for Phase 1, as their gate conditions — CRA enforcement, ENISA PQC guidelines, and Simpl-Open launch — are prerequisites for the majority of downstream priorities. A single cross-cutting bottleneck dominates Phase 2: RISC-V hardware scaling under the EU Chips Act, which constitutes the shared gate condition for P1, P2, P3, and P8 Phase 2→3 transitions simultaneously.

Capability maturity is tracked using a dual-metric framework: **Operational Deployment Phase (ODP)** for market integration and **Strategic Sovereignty Level (SSL)** for foreign technology independence, across three phases: *Foundation & Compliance* (0–5 years), *Federation & Standardization* (5–10 years), and *Autonomy & Leadership* (10–15 years).

Phase 1 Recommendations (2025–2030): Foundation, Compliance, and Applied Engineering

Phase 1 is the applied engineering and compliance activation phase. Research is primarily integration research — deploying near-TRL solutions and operationalizing obligations under the CRA [2], NIS2 [1], DORA [12], and AI Act [3]. Policy and Compliance is the dominant stream: the majority

of Phase 1→2 gate conditions are regulatory enforcement events. Four priorities — P3 (zero-energy PhySec), P6 (autonomous AI agents), P8 (FHE at scale), and P9 (Cognitive Twin orchestration) — remain in Applied R&D in this phase; their Phase 1 bullets are therefore research-oriented rather than deployment-oriented.

Technical R&I Recommendations

- **P1, P8 — Deploy hardware attestation stacks and TEE environments on critical nodes.** Deploy remote attestation pipelines based on TPM 2.0 [13] and DICE protocols for critical cloud-edge nodes. Implement RISC-V secure enclave pilots (Keystone [14], OpenTitan [15]) alongside TEE deployments (Intel TDX, AMD SEV-SNP, ARM CCA) for regulated cloud workloads, maintaining multi-vendor diversity to prevent single-vendor side-channel exposure. These deployments establish the hardware trust substrate for both the **P1 EUCC ‘High’ gate** and the **P8 EUCS Certification Rollout gate**.
- **P2 — Implement and validate hybrid PQC schemes on resource-constrained targets.** Implement hybrid cryptographic stacks combining ECDH-P384 with ML-KEM (FIPS 203 [9]) and ML-DSA (FIPS 204 [10]) for cloud and edge environments. Validate performance on ARM Cortex-M4 and 8/16-bit microcontrollers against NIST IR 8413 [16] reference measurements, providing the quantitative baseline for the **P2 ENISA PQC Guidelines gate**.
- **P3 — Validate lightweight PhySec primitives on constrained IoT hardware (Applied R&D).** Validate group-based authentication and RF fingerprinting using Channel State Information (CSI) and RSSI on EU-manufactured constrained IoT hardware, establishing PhySec baselines for anomaly detection ahead of the **P3 NIST Lightweight PQC Standards gate**.
- **P5 — Deploy Runtime SBOM tooling and signed OTA update pipelines.** Deploy Runtime SBOM tooling with integrity attestation for containerized microservices and cryptographically signed OTA update pipelines for field-deployed edge nodes (smart grids, O-RAN radios, industrial gateways), meeting the **P5 CRA supply chain enforcement gate**.
- **P6, P10 — Integrate LLM-assisted SOC copilots with NWDAF telemetry and deploy federated SIEMs (Applied R&D).** Integrate LLM-assisted analyst tools with 3GPP NWDAF (TS 23.288 [17]) telemetry, and deploy federated SIEM infrastructure with STIX/TAXII CTI sharing aligned with MITRE ATT&CK. Validate ML-based IoC correlation for APT dwell-time reduction. Autonomous AI agents target the **P6 EU AI Act enforcement gate**; federated SIEMs meet the **P10 CTI Sharing Protocols gate**.
- **P9 — Implement soft-state protocols for Island Mode piloting.** Implement soft-state session protocols and time-bounded cryptographic credential caches enabling validated autonomous edge operation during WAN disconnection, establishing the fallback piloting baseline for the **P9 NIS2 and DORA gate**.

Policy and Compliance Recommendations

- **P1 — Enforce CRA hardware security requirements.** Issue CRA [2] implementing regulations specifying minimum hardware security requirements for products with digital elements entering the EU market and mandate compliance attestation for critical infrastructure operators by 2026. This is the **P1 Phase 1→2 gate condition**.

- **P2 — Publish ENISA sector-specific PQC migration guidance.** Publish ENISA PQC migration guidance mapping FIPS 203/204/205 to NIS2 [1] Article 21(2)(e) cryptographic policy obligations, with sector-specific migration timelines published by 2026. This activates the **P2 ENISA PQC Guidelines gate**.
- **P4, P5 — Mandate Simpl MIM adoption and machine- readable SBOMs as EU funding conditions.** Mandate security MIM compliance as a contractual condition for Digital Europe Programme Data Space funding, and require machine-readable SBOM formats for all digital products under CRA implementing regulations. Establish an SME compliance support scheme through ECCC, activating the **P4 Simpl-Open gate** and the **P5 CRA supply chain gate** simultaneously.
- **P8 — Issue EUCS ‘High+’ guidance mandating confidential computing.** Issue EUCS [4] ‘High+’ implementing guidance requiring TEE-based confidential computing for sensitive cloud workload categories, and align GDPR [18] Article 25 enforcement guidance with confidential computing technical standards. This activates the **P8 EUCS Certification Rollout gate**.
- **P9 — Issue NIS2 and DORA business continuity guidance for edge infrastructure.** Issue NIS2 Article 21 [1] implementing guidance requiring business continuity testing for edge-dependent critical infrastructure, and integrate Island Mode assessment into DORA [12] ICT risk management reviews. These jointly constitute the **P9 NIS2 and DORA gate**.
- **P10 — Establish legally harmonized cross-border CTI sharing agreements.** Establish legally harmonized CTI sharing agreements addressing jurisdictional barriers, and mandate MITRE ATT&CK integration for all EU-funded SIEM deployments to ensure semantic interoperability of IoC data. This is the **P10 CTI Sharing Protocols gate**.

Funding and Ecosystem Recommendations

- **P1, P8 — Fund RISC-V secure element pilots and sovereign silicon feasibility studies.** Fund 3–5 RISC-V secure element pilot projects targeting EUCC ‘High’ certification under Horizon Europe Cluster 3, with a minimum allocation of EUR 50M for sovereign secure silicon R&D. Allocate EU Chips Act funding to fabrication line feasibility studies as the industrial precursor to Phase 2 volume production.
- **P2, P3 — Fund lightweight PQC and PhySec R&D through ECCC and Digital Europe.** Fund ECCC-led research into lightweight PQC optimization for 8/16-bit microcontrollers and ETSI SmartBAN standardization for WBAN security, creating the experimental base for Phase 2 hardware-software co-design.
- **P4, P7 — Fund Simpl governance and IPCEI-CIS 5G corridor deployments.** Fund the Simpl governance board security workstream through Digital Europe Programme to publish initial security MIM definitions by 2026, and activate IPCEI-CIS cross-border 5G corridor infrastructure in at least three corridors by 2027.
- **P6, P10 — Activate EuroHPC AI Factories and fund the Pan-European CTI platform.** Activate EuroHPC AI Factory compute allocations for sovereign security AI model training, and fund a Pan-European CTI platform succeeding CONCORDIA [8] with legally harmonized intelligence sharing spanning all 27 Member States.

Success Metrics for 2030

By the end of Phase 1, Europe should achieve:

- **Hardware Sovereignty:** At least 30% of critical infrastructure edge nodes utilizing EU-designed, EUCC ‘High’ certified secure elements; first sovereign RISC-V secure element in industrial production.
- **Quantum Readiness:** 100% of new critical IoT deployments supporting hybrid (classical + PQC) schemes; complete cryptographic inventory of 80%+ legacy OT/IoT systems.
- **Federated Trust:** Majority of Common European Data Spaces operating on Simpl-based security interoperability with standardized MIMs adopted as CEN/CENELEC standards.
- **Certification Maturity:** Operational continuous certification frameworks for at least two major Data Spaces with real-time EUCC/EUCC tracking; point-in-time audit model deprecated in pilot sectors.
- **Cognitive SecOps Readiness:** LLM-assisted SOC tools deployed in at least five Member State national cybersecurity centres; NWDAF telemetry pipelines feeding centralized AI analytics for anomaly detection.
- **Supply Chain Transparency:** Runtime SBOM frameworks mandated for all new EU-funded cloud/edge deployments; Digital Product Passport pilots operational in the telecom and industrial IoT sectors.

Phase 2 Recommendations (2030–2035): Applied and Industrial R&D, Federation, Standardization

Phase 2 is the applied and industrial R&D phase. Gate conditions shift from regulatory triggers to standardization milestones and sovereign capability consolidation. The dominant shared dependency across P1, P2, P3, and P8 is RISC-V hardware scaling under the EU Chips Act — the single most consequential cross-cutting bottleneck of the roadmap, whose resolution enables four simultaneous Phase 2→3 transitions. Technical R&I is the dominant stream.

Technical R&I Recommendations

- **P1, P8 — Scale RISC-V secure elements to industrial fabrication and benchmark against incumbents.** Scale RISC-V secure element designs from TRL 6–7 to industrial-volume fabrication under EU Chips Act production lines. Benchmark EUCC ‘High’-certified EU secure elements against non-EU incumbents (Intel, NXP, Infineon) on performance, power, and tamper resistance. This milestone simultaneously constitutes the **P1 EU Chips Act gate** and the **P8 RISC-V Hardware Scaling gate**, confirming sovereign silicon as the primary shared bottleneck.
- **P2 — Develop crypto-agility middleware and scale hybrid PQC to Industrial IoT.** Develop and standardize crypto-agility middleware enabling OTA algorithm migration from classical to hybrid ML-KEM/ML-DSA without hardware replacement. Scale hybrid PQC across Industrial IoT, energy grid, and financial infrastructure, benchmarking coverage against the **P2 NIST/ETSI finalization gate**.

- **P3 — Develop hardware-software co-design for neuromorphic and PIM extreme-edge security.** Develop hardware-software co-design implementations embedding lightweight PQC into EU RISC-V profiles for WBAN and extreme-edge contexts. Research neuromorphic and Processing-in-Memory (PIM) architectures for energy-constrained cryptographic processing. Standardize ETSI SmartBAN security extensions, meeting the **P3 Hardware-Software Co-Design gate**.
- **P4 — Develop automated policy translation engines and federate IAM across Data Spaces.** Develop automated policy translation engines converting provider-specific security policies into XACML and OSCAL formats. Federate IAM bridges using W3C Verifiable Credentials, scaling Simpl-based security to a majority of EU Common Data Spaces and advancing toward the **P4 Security MIMs CEN/CENELEC gate**.
- **P5 — Develop dynamic Runtime SBOMs and research ZKP supply chain attestation.** Develop dynamic Runtime SBOM tooling for real-time active component tracking across multi-provider deployments. Research and prototype ZKP mechanisms for IP-protected component verification, and standardize Digital Product Passport data models for ICT components, meeting the **P5 Runtime SBOMs and DPP gate**.
- **P6, P10 — Develop sovereign edge LLMs, standardize AI Interconnect APIs, and scale UEBA with MTD pilots.** Develop sovereign European edge LLMs and SLMs for security classification at EuroHPC AI Factories. Standardize AI Interconnect APIs extending ETSI GS ZSM 009 [19] to cybersecurity-specific agent interoperability, meeting the **P6 AI Interconnect API gate**. Concurrently, scale UEBA behavioral baselines using harmonized STIX/TAXII ontologies across all 27 Member States and pilot Moving Target Defense (MTD) in at least three Member State critical infrastructure sectors, meeting the **P10 LLMSecOps and UEBA gate**.
- **P7 — Develop inter-operator east-west security interfaces and pilot Data Visa frameworks.** Develop and standardize inter-operator east-west security interfaces for federated Zero-Trust enforcement across national operator boundaries. Research and pilot Data Visa legal-technical mechanisms for cross-border AI workload migration validated in at least two cross-border corridors by 2032, meeting the **P7 East-West Interfaces gate**.
- **P8 — Develop multi-party confidential AI pipelines and research FHE performance.** Develop multi-party SMPC and federated learning pipelines across multi-provider regulated sectors (health, finance). Research FHE performance optimization targeting non-real-time production analytics viability, establishing the trajectory toward the **P8 FHE Maturation gate**.
- **P9 — Develop and standardize 3GPP/ETSI fallback APIs for autonomous Island Mode.** Develop 3GPP/ETSI fallback API specifications for autonomous Island Mode activation and credential caching. Research AI-driven graceful degradation mechanisms and dynamic spectrum sharing for emergency edge roaming, meeting the **P9 3GPP/ETSI Fallback API gate**.

Policy and Compliance Recommendations

- **P1, P2 — Mandate EUCC ‘High’ in procurement and publish EU hybrid PQC scheme profiles.** Mandate EUCC ‘High’ [11] certification for all new critical infrastructure ICT procurement across all 27 Member States from 2031. Concurrently, publish EU-specific hybrid PQC scheme profiles (classical + ML-KEM; classical + ML-DSA) through ETSI TC CYBER by 2031 and mandate classical-only cryptography deprecation for high-security EU government contexts by 2033.
- **P6 — Regulate autonomous AI security agents under EU AI Act Article 9.** Regulate autonomous AI security systems under AI Act [3] Article 9 high-risk requirements, mandating explainability, auditability, and human override for all autonomous containment decisions. Publish ECCC AI Trustworthiness guidance for LLMSecOps deployments in critical infrastructure.
- **P7, P9 — Ratify Data Visa instruments and mandate Island Mode certification.** Ratify Data Visa legal instruments enabling cross-border AI workload migration through DG CONNECT and DG JUST by 2032. Mandate Island Mode capability certification in NIS2 business continuity assessments for edge-dependent critical infrastructure from 2032.

Funding and Ecosystem Recommendations

- **P1, P2, P8 — Direct EU Chips Act investment toward unified RISC-V secure and PQC-native silicon.** Direct EU Chips Act fabrication investment toward unified RISC-V secure elements integrating TEE and PQC hardware acceleration in volume production by 2032. Fund Chips JU co-design programmes aligning sovereign silicon profiles with simultaneous EUCC ‘High’ and EUCS ‘High+’ certification pathways.
- **P3, P6 — Fund neuromorphic/PIM research and scale EuroHPC AI Factory sovereign security AI.** Fund Horizon Europe Cluster 4 research targeting neuromorphic and PIM architectures for energy-constrained security. Scale EuroHPC AI Factory allocations for sovereign LLM/SLM training, with benchmark evaluation against non-EU foundation models on security-relevant classification tasks.
- **P4, P7, P10 — Fund Simpl MIM standardization, CAMARA inter-operator profiles, and CTI platform operations.** Fund CEN/CENELEC JTC 13 standardization of Simpl security MIMs and CAMARA API security profile development through IPCEI-CIS. Scale the Pan-European CTI platform with UEBA behavioral baseline harmonization covering all Member States.

Success Metrics for 2035

By the end of Phase 2, Europe should achieve:

- **Quantum-Resilient Infrastructure:** 100% of EU public-sector ICT using hybrid or full PQC schemes; first “Quantum-Safe Data Space” fully operational; classical-only cryptography deprecated in all high-security EU government systems.
- **Sovereign Silicon Dominance:** 60%+ EU market share for secure silicon in critical edge infrastructure; EUCC ‘High’ certification mandatory for regulated sector ICT procurement across all 27 Member States.

- **Federated NaaS Security:** Pan-European NaaS security framework operational across at least 10 telecom operators; “Data Visa” cross-border AI workload migration demonstrated in production at corridor scale; 70%+ Common Data Spaces using automated Simpl-based policy federation.
- **Autonomous AI SecOps:** Edge-native Guardian nodes autonomously executing zero-trust reconfigurations deployed in at least five Member State critical infrastructure sectors; sovereign EU edge LLMs/SLMs operationally replacing non-EU foundation models in security analytics.
- **Confidential Computing at Scale:** Multi-party privacy-preserving AI pipelines operational in at least three cross-border regulated sectors; RISC-V TEEs scaling toward European silicon self-sufficiency for enclave hardware.
- **APT Dwell Time Reduction:** Mean APT dwell time reduced by 70% compared to Phase 1 baseline through AI-driven behavioral hunting; Pan-European CTI sharing platform operational with legally harmonized data exchange agreements covering all 27 Member States.

Phase 3 Recommendations (2035–2040): Frontier Research and Sovereign Operational Leadership

Phase 3 carries two simultaneous trajectories. For capabilities industrialized in Phase 2 — P1, P4, P7 — this phase represents full operational maturity and global standard-setting; dominant verbs are *operationalize*, *establish*, and *lead*. For capabilities still in advanced R&D — P3, P6, P8, P9, P10 — Phase 3 is where the hardest scientific problems are actively solved; dominant verbs are *pioneer*, *advance*, and *research*. Funding and Ecosystem is the dominant stream, as sovereign industrial self-sufficiency and global standards export require sustained investment beyond individual R&I projects.

Technical R&I Recommendations

- **P1, P2 — Operationalize PQC-native RISC-V across all EU tiers and establish global PQC standards.** Operationalize PQC-native RISC-V silicon across all EU Cloud-Edge-IoT tiers by 2036. Retire all hybrid classical/PQC transitional schemes from sovereign EU systems, meeting the **P2 PQC-native RISC-V gate**. Establish ETSI/CEN/CENELEC-led PQC migration standards as the global reference baseline through ISO and ITU-T, meeting the **P1 EU Chips Act fabrication gate**.
- **P3 — Pioneer zero-energy air interfaces with native ISAC security primitives for 6G.** Pioneer zero-energy air interfaces embedding ISAC security primitives as mandatory 6G air interface features, aligned with SNS JU SRIA [5]. Research ambient RF energy harvesting for battery-less node authentication, establishing ubiquitous ISAC as the convergent security-communications substrate and meeting the **P3 Zero-Energy Air Interfaces and ISAC gate**.
- **P5 — Operationalize ZKP-based supply chain attestation at continental scale.** Operationalize ZKP-based attestation as the standard cryptographic mechanism for IP-protected component verification across all EU critical infrastructure. Integrate automated supply chain compromise indicators into the Pan-European CTI platform as a real-time feed, meeting the **P5 ZKPs and Pan-European threat intelligence gate**.

- **P6 — Pioneer self-evolving AI swarm architectures for zero-touch sovereign cyber defense.** Pioneer decentralized self-evolving AI swarm architectures for zero-touch cyber defense, with governance frameworks for AI trustworthiness and algorithmic accountability satisfying AI Act [3] Article 9 high-risk system requirements. Establish EuroHPC as the federated training infrastructure for continuously self-improving sovereign security AI, meeting the **P6 Self-Evolving AI Swarm Protocols gate**.
- **P8 — Operationalize FHE for real-time analytics and pioneer SMPC standardization.** Operationalize FHE for real-time production analytics by 2037–2038, completing the trajectory from isolated TEE enclaves to a ubiquitous confidential continuum from battery-less IoT to HPC, meeting the **P8 FHE Maturation gate**. Pioneer SMPC standardization through CEN/CENELEC for cross-border federated AI in regulated sectors, ensuring no raw sensitive data crosses sovereign boundaries.
- **P9 — Pioneer Cognitive Twin E2E orchestration for autonomous cyber-physical recovery.** Pioneer Cognitive Twin architectures for E2E autonomous cyber-physical recovery, advancing Phase 2 digital twins to operational autonomous management. Research bio-inspired elasticity protocols through ETSI ENI, establishing vendor-interoperable autonomous recovery as the normative resilience posture and meeting the **P9 Cognitive Twin gate**.
- **P10 — Operationalize AI-driven MTD and pioneer proactive counter-APT swarm intelligence.** Operationalize AI-driven Moving Target Defense continuously shuffling network resources, IP addresses, and software diversity without human intervention, meeting the **P10 Adaptive AI Swarms and Zero-Touch Automation gate**. Pioneer proactive counter-adversarial intelligence generation disrupting APT operations before objectives are achieved. All autonomous MTD decisions must satisfy EU AI Act explainability requirements, establishing a hard prerequisite dependency on the **P6 AI governance gate**.

Policy and Compliance Recommendations

- **P4, P7 — Enact EU Cloud Rulebook and operationalize unified EU computing jurisdiction.** Enact the EU Cloud Rulebook as binding law by 2037–2038, mandating provider-agnostic Zero-Trust federation for all EU-funded digital infrastructure. Operationalize the unified EU computing jurisdiction for AI workload governance, jointly meeting the **P4 EU Cloud Rulebook gate** and the **P7 “Schengen for Cloud” gate**.
- **P1, P2, P5 — Mandate PQC-by-default, EUCC ‘High’, and full supply chain blocking for all new EU ICT products.** Mandate PQC-by-default and EUCC ‘High’ for all new EU ICT products after 2036, and mandate full automated blocking of unverified hardware, OS, or software components from EU critical infrastructure, transitioning P5 from partial to full restriction.
- **P6, P10 — Enforce AI trustworthiness requirements prior to full swarm deployment.** Enforce AI trustworthiness and algorithmic accountability under AI Act Article 9 for all autonomous security agents before AI swarm architectures reach operational scale. This enforcement is a named prerequisite gate for both P6 and P10 Phase 3 deployment.

Funding and Ecosystem Recommendations

- **P1, P2 — EU Chips Act sovereign fabrication to meet 80%+ of European secure silicon demand by 2039.** Sustain EU Chips Act investment to reach 80%+ domestic fabrication coverage of European secure silicon demand by 2039. Position open-source RISC-V EUCC ‘High’ architectures as the global reference implementation for hardware roots of trust, generating measurable export value for the European semiconductor industry.
- **P3, P6, P9, P10 — Fund Horizon Europe frontier research for zero-energy ISAC, AI swarms, Cognitive Twins, and autonomous MTD.** Fund Horizon Europe frontier research programmes (ERC Advanced Grants, SNS JU Phase 3, Future and Emerging Technologies) for zero-energy ISAC, self-evolving AI swarm governance, Cognitive Twin orchestration, and autonomous MTD. Establish European research leadership in these domains as the precondition for global standard-setting in post-6G and neuromorphic computing security.
- **P4, P7 — Export European cybersecurity standards as global reference frameworks.** Actively promote EU cybersecurity standards — EUCS, EUCC, security MIMs, PQC migration frameworks, and continuous certification lifecycle models — through the EU-US Trade and Technology Council, EU-Global Gateway, ISO, and ITU-T. Position European frameworks as the global reference for trustworthy, sovereign, and sustainable computing security, with export value assessed in the 2040 roadmap completion review by ECCC and ENISA.
- **P8 — Fund ECCC FHE optimisation and CEN/CENELEC SMPC standardization.** Fund an ECCC-led FHE performance optimisation programme targeting real-time edge analytics viability by 2037–2038. Fund CEN/CENELEC SMPC standardization for regulated cross-border AI, ensuring privacy-preserving federated computation becomes a binding interoperability requirement.

Success Metrics for 2040

By the end of Phase 3, Europe should achieve:

- **Full Quantum-Native Continuum:** 100% of EU Cloud-Edge-IoT critical infrastructure natively running PQC; EU PQC and QKD standards adopted as ISO/ITU-T global baseline; no operational classical-only cryptography in sovereign EU systems.
- **Sovereign Silicon Self-Sufficiency:** 80%+ of European secure silicon demand met by EU Chips Act fabrication capacity; RISC-V EUCC ‘High’ the default hardware foundation for all new EU ICT products.
- **Unified European Digital Jurisdiction:** EU Cloud Rulebook fully enforced; 100% of Common Data Spaces operating on provider-agnostic zero-trust architectures; “Schengen for Cloud-Edge” legally operational across all 27 Member States.
- **Autonomous AI Security Fabric:** Self-evolving AI swarm defense deployed across the EU continuum; all autonomous security agents compliant with EU AI Act Trustworthiness and Accountability requirements; Security-by-Design the normative default — manual security configuration obsolete.

- **Ubiquitous Confidential Computing:** FHE operationalized for real-time production analytics; SMPC standardized for cross-border federated AI in all regulated sectors; end-to-end encrypted continuum from battery-less IoT to HPC without raw sensitive data leaving sovereign boundaries.
- **Full Counter-APT Resilience:** AI-driven MTD continuously disrupting APT reconnaissance in real time; Pan-European CTI platform providing proactive counter-adversarial intelligence; mean APT dwell time reduced to near-zero for known attack pattern families.
- **Global Leadership:** European cybersecurity frameworks serving as the global reference implementation; EU cybersecurity technology export value exceeding pre-roadmap baseline by a measurable factor, as assessed by ECCC and ENISA in the 2040 roadmap completion review.

10 Discussion

The proposed methodology bridges a gap between architecture-centric security analysis and policy-oriented R&I prioritization. Its main strength lies in explicit traceability from candidate capability to executive recommendation, using structured filters, hybrid profiling dimensions, and phase-based progression logic.

From an IEEE Access perspective, the framework can be interpreted not only as a strategic roadmap artifact but also as a reusable methodological contribution for evaluating sovereign cybersecurity capabilities in heterogeneous computing continua.

11 Conclusion

This paper presented a structured and traceable methodology for cybersecurity prioritization in the Cloud–Edge–IoT continuum. By integrating architectural modeling, multi-dimensional prioritization, and phased progression analysis, the framework enables evidence-based decision-making aligned with European digital sovereignty objectives.

Future work includes quantitative validation, integration with real deployment data, and extension toward automated policy-driven orchestration.

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